# Geometric Modeling
# Assignment 4: Mesh Parametrization

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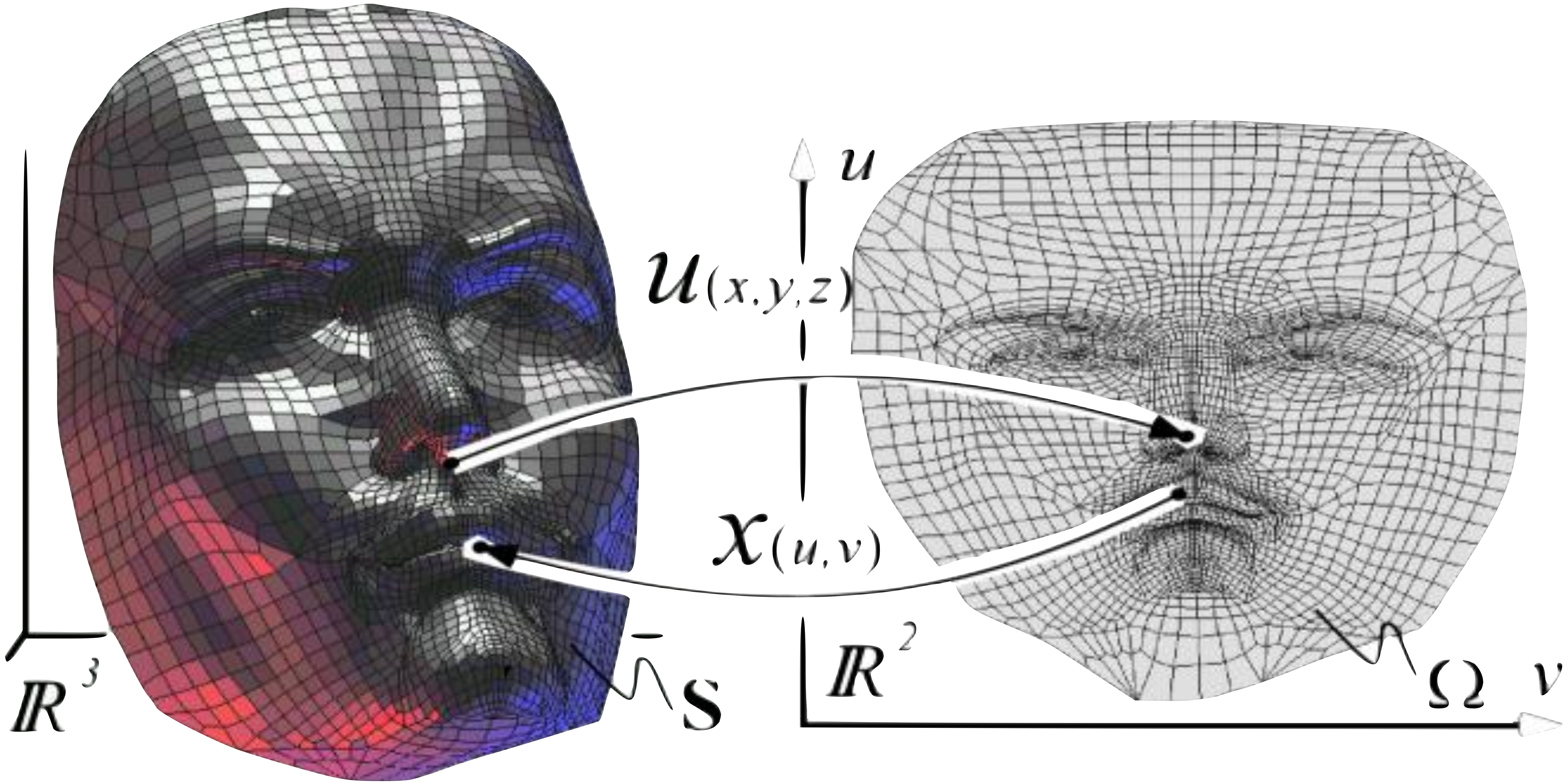
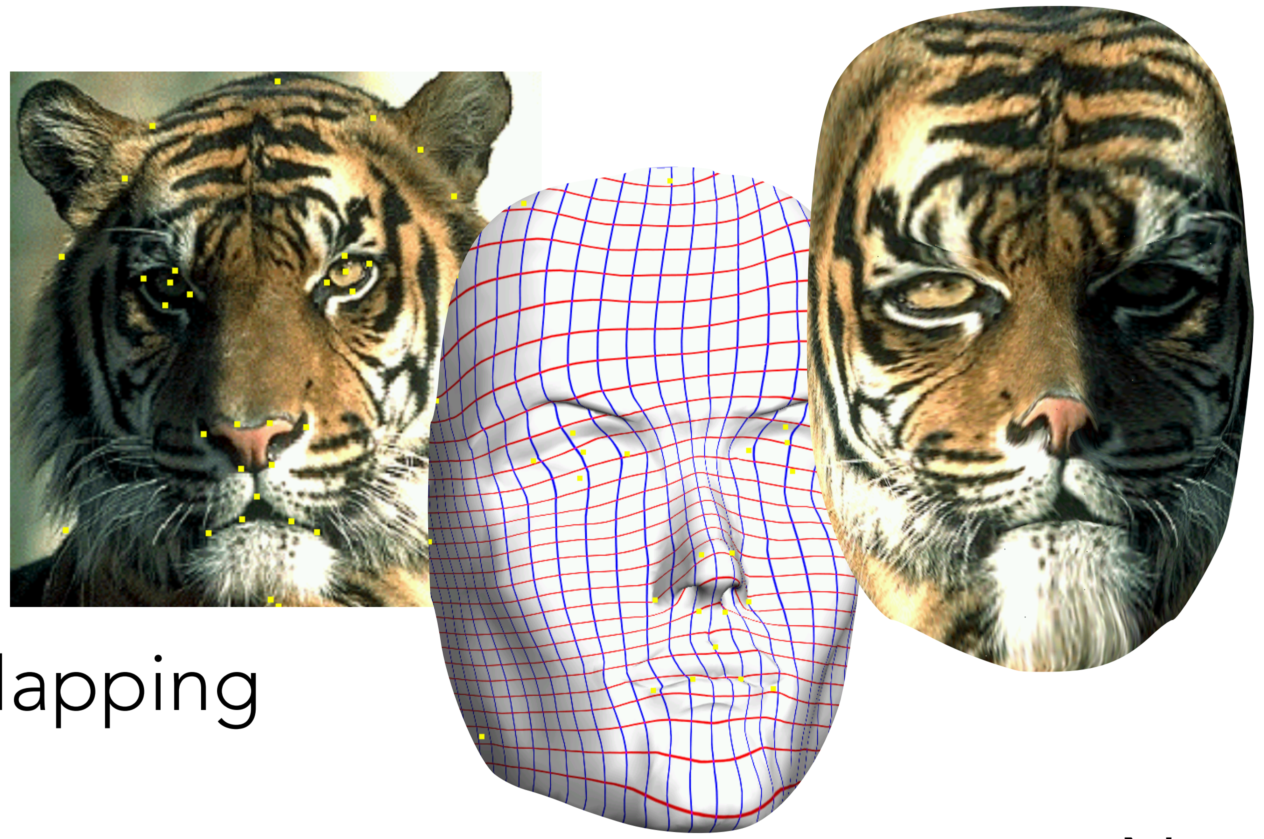
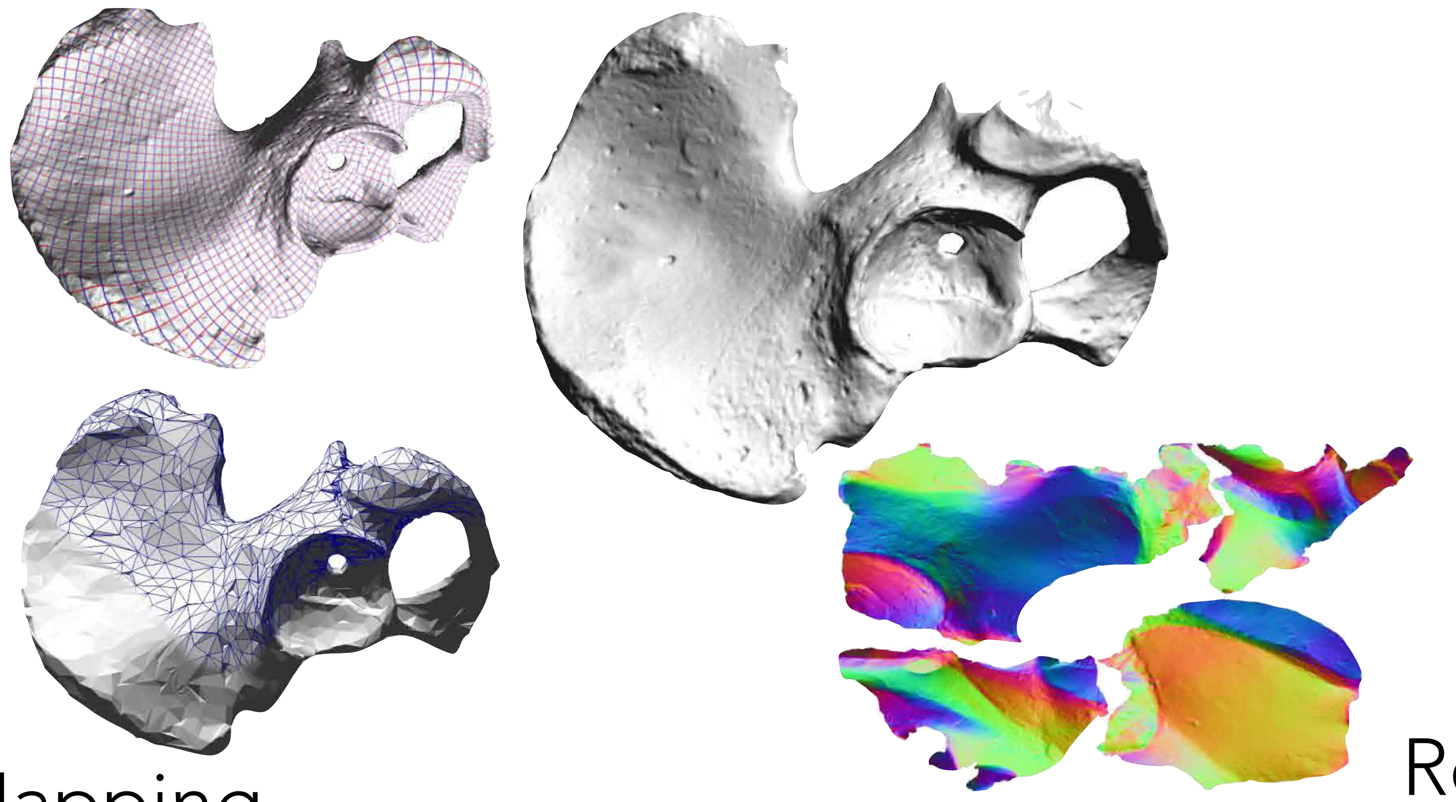
# Mesh Parametrization



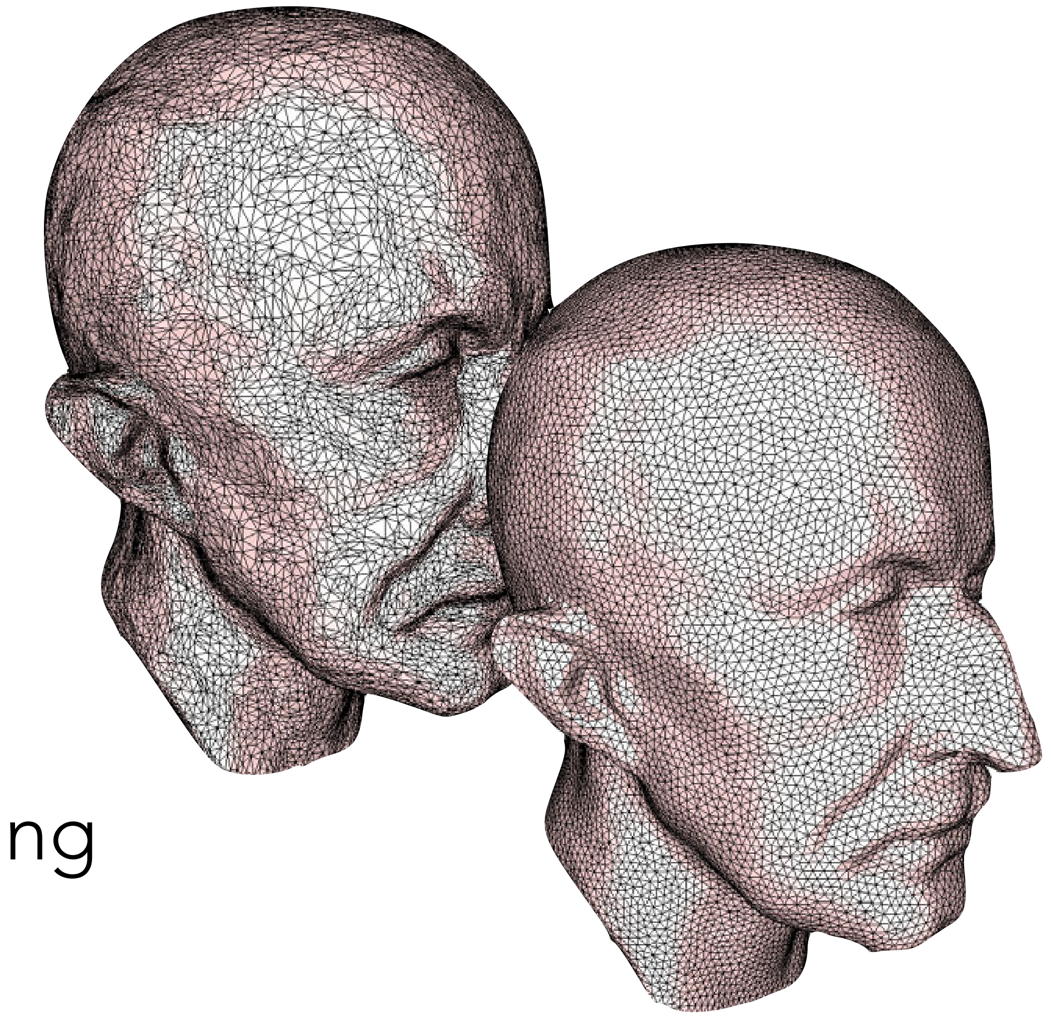
# Parametrization Applications



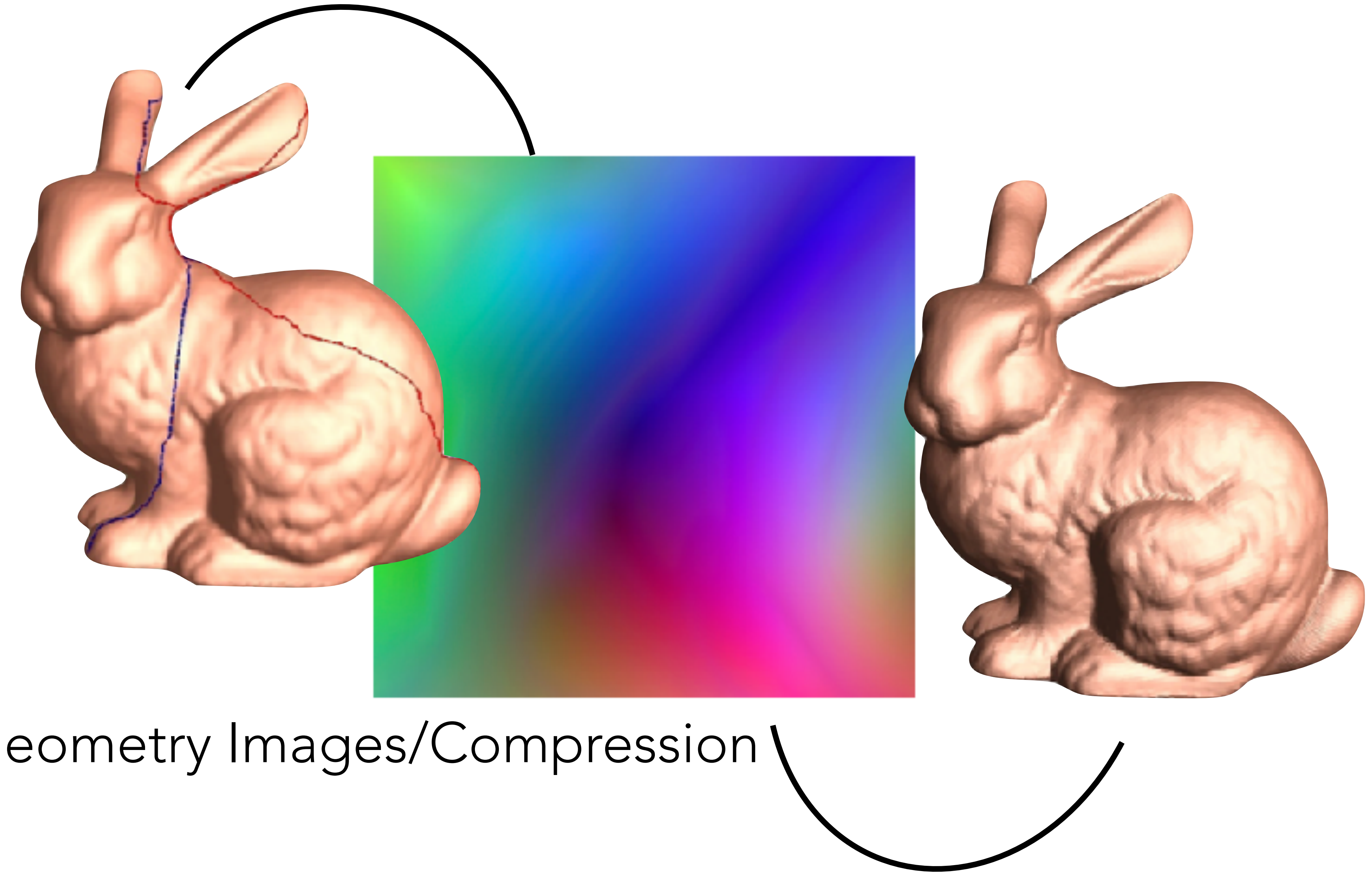
Texture Mapping



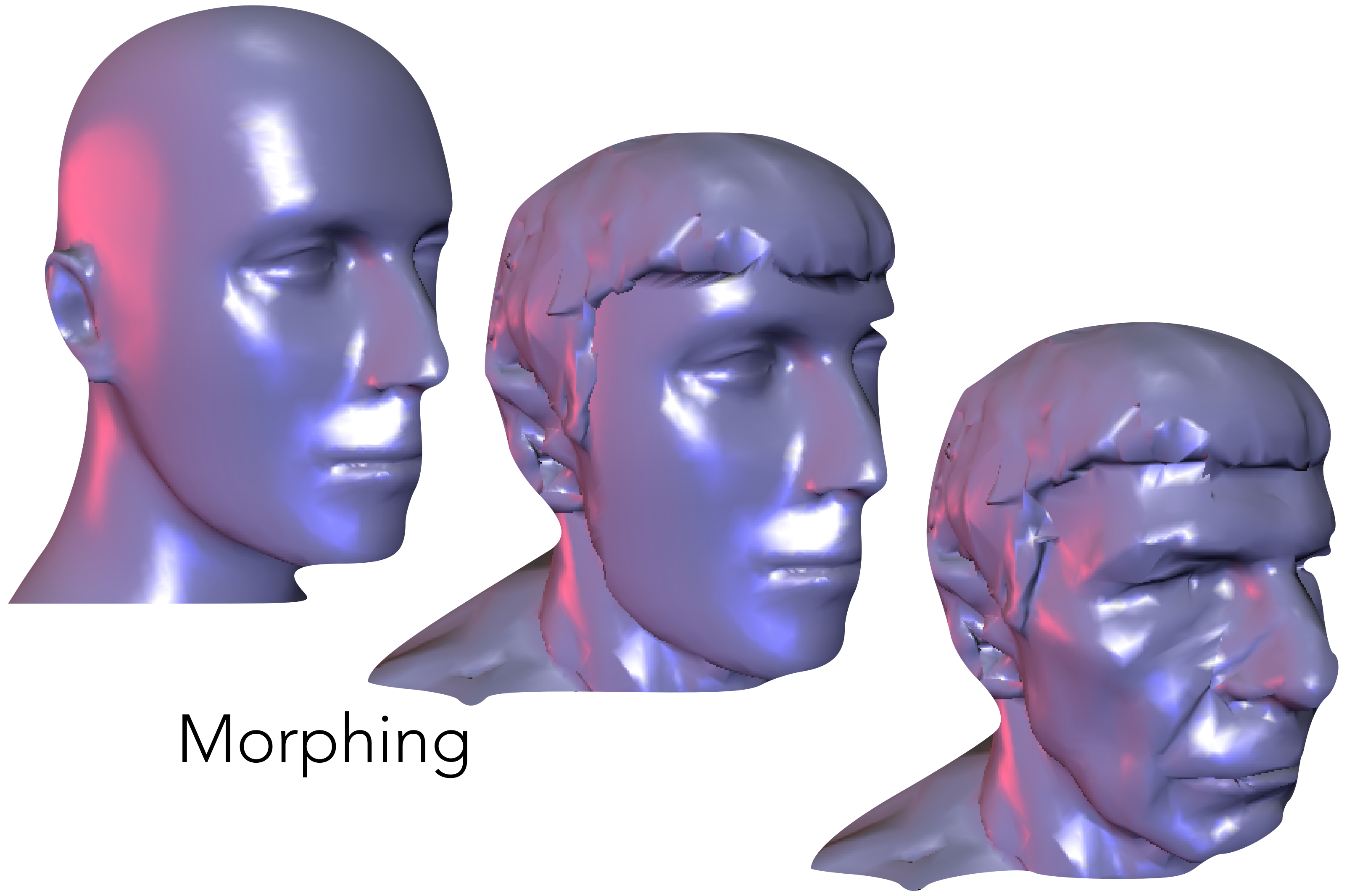
Normal/Bump Mapping



Remeshing



Geometry Images/Compression



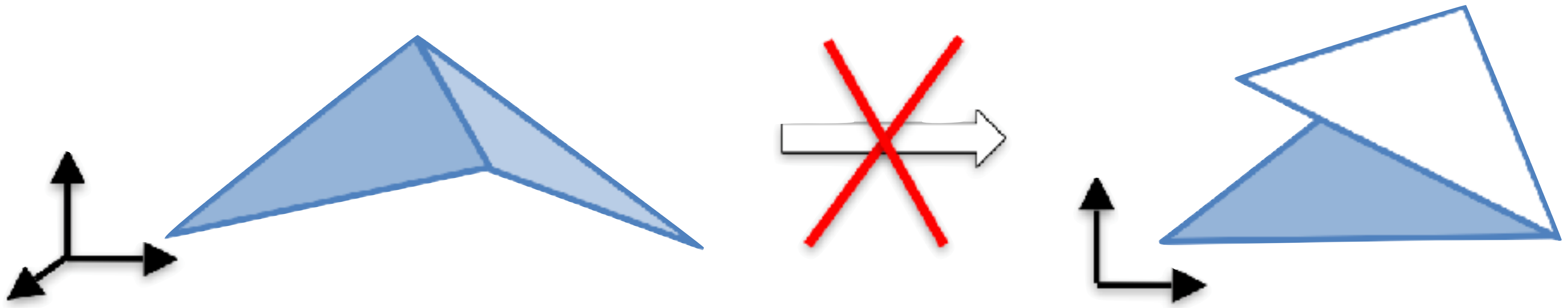
Morphing

- Detail Transfer
- Mesh Completion
- Editing
- Surface Fitting

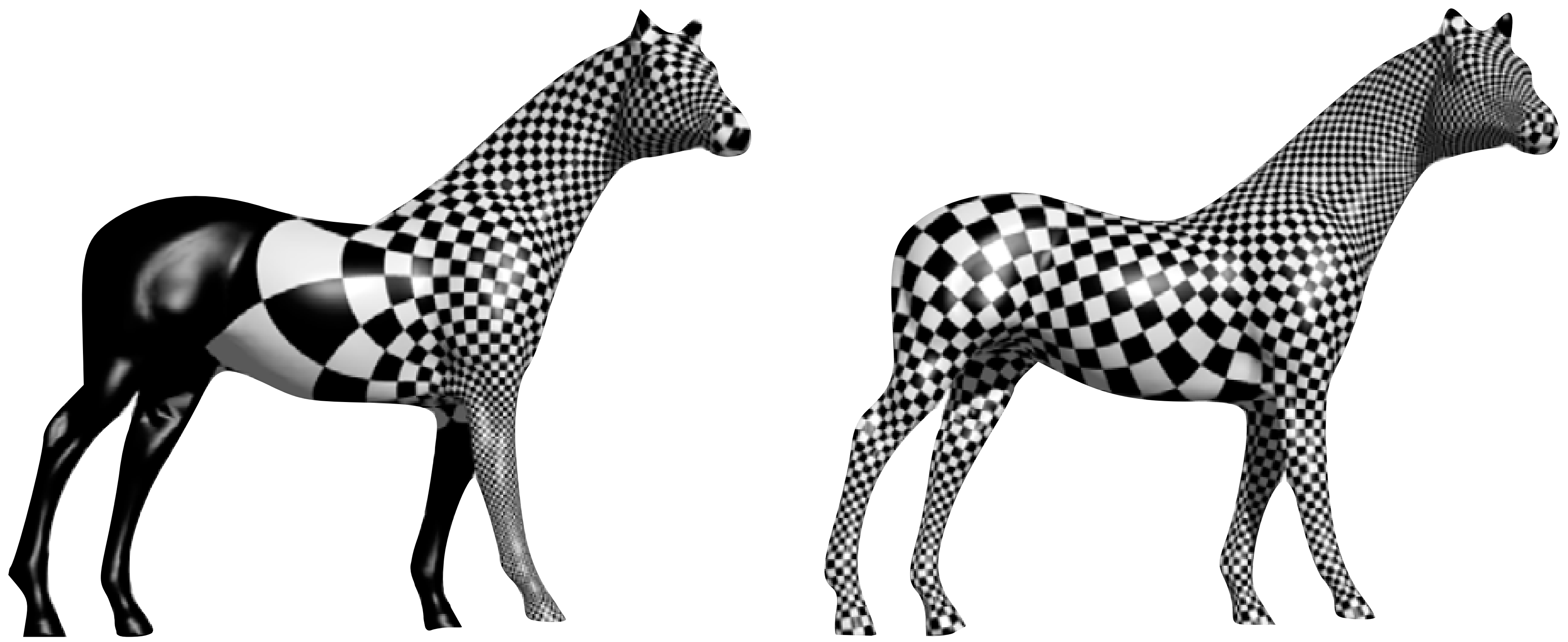
...

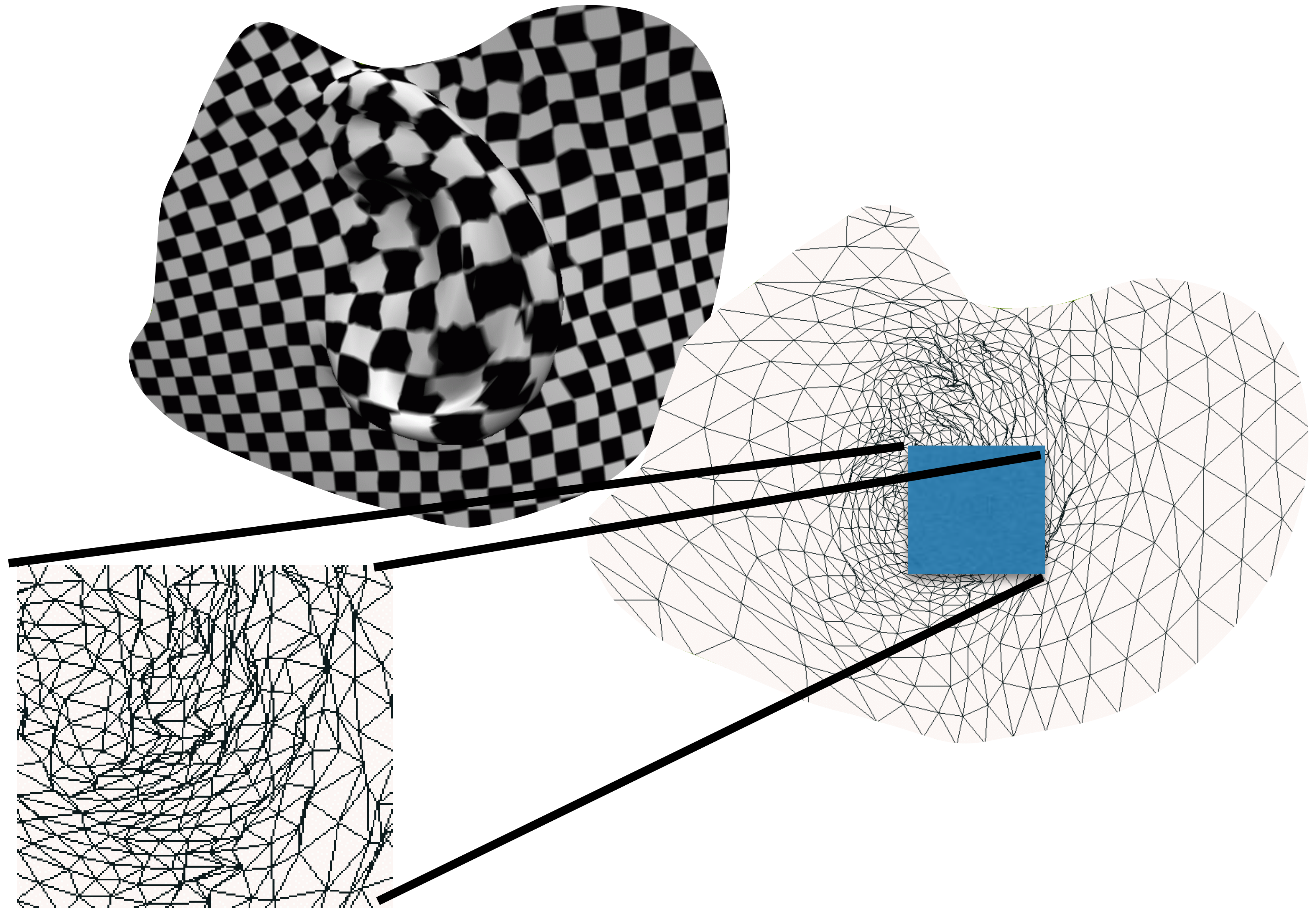
# Desirable Properties

- Bijective (invertible): avoid flips/overlaps



- Minimal distortion
  - Area
  - Angle





- Efficiently computable

# Parametrization Approaches

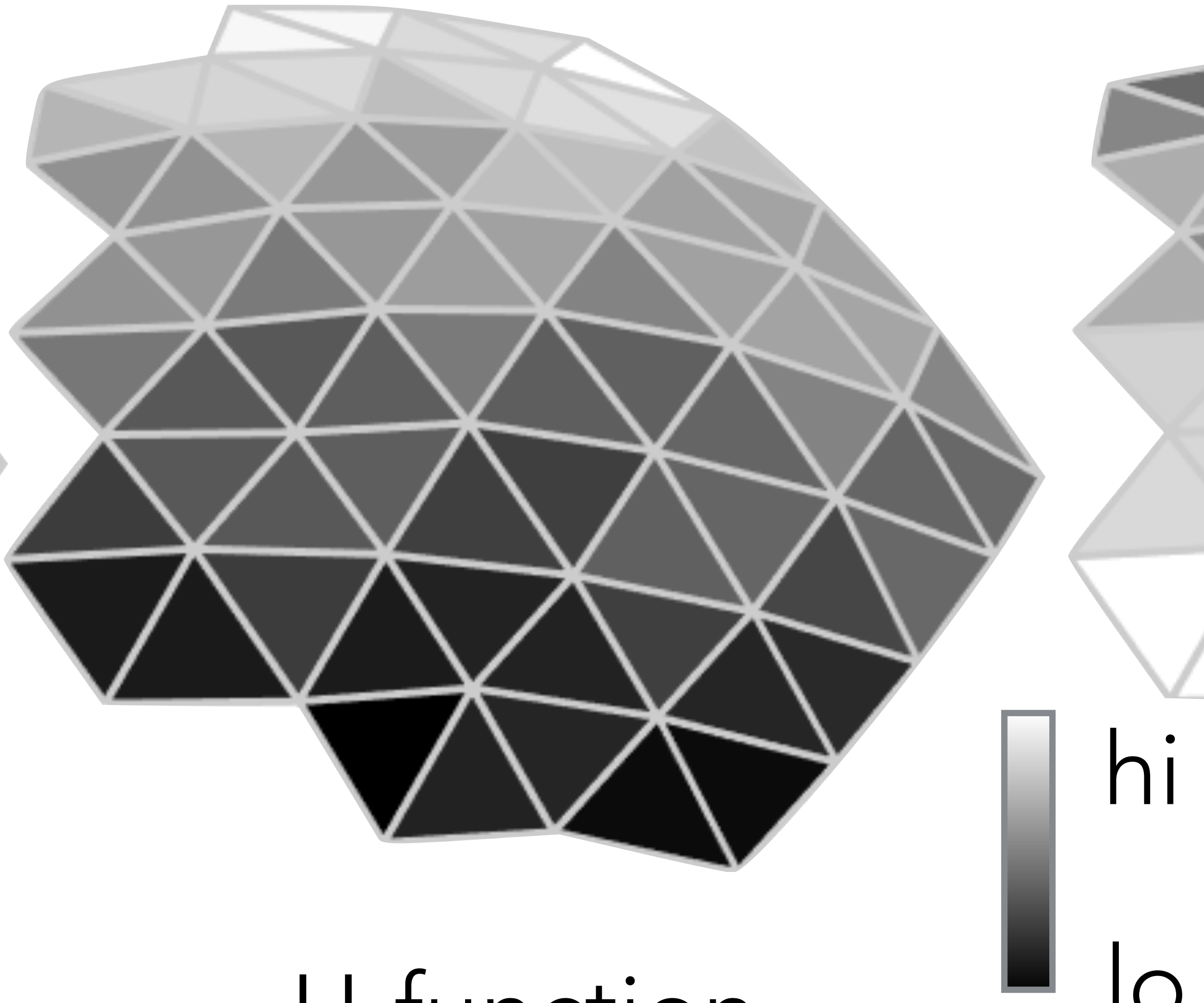
- This assignment will consider 3 approaches
  - Least-Squares Conformal (LSCM)
  - Harmonic
  - Field-Guided
- LSCM, Harmonic implemented in libigl

# Formulation: Design Scalar Fields

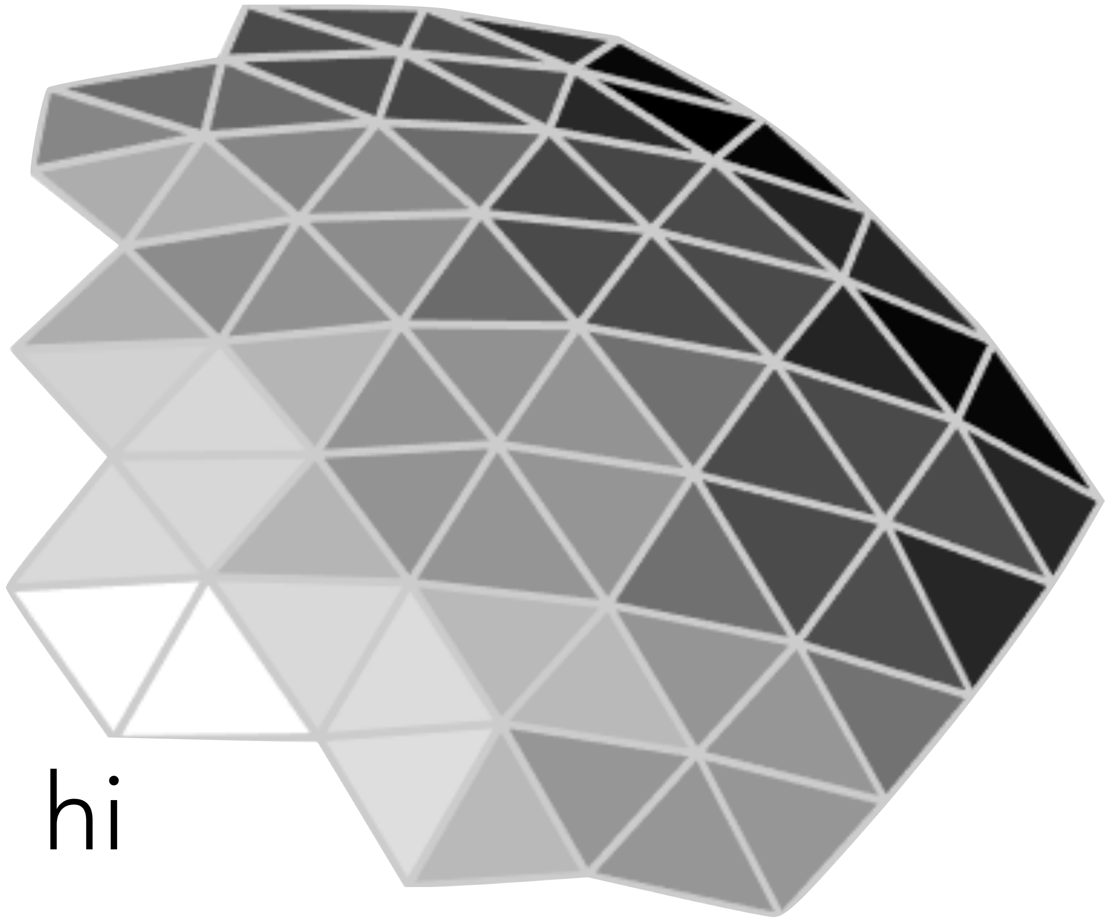
- Compute two scalar fields (u, v) giving the plane coordinates of each mesh point



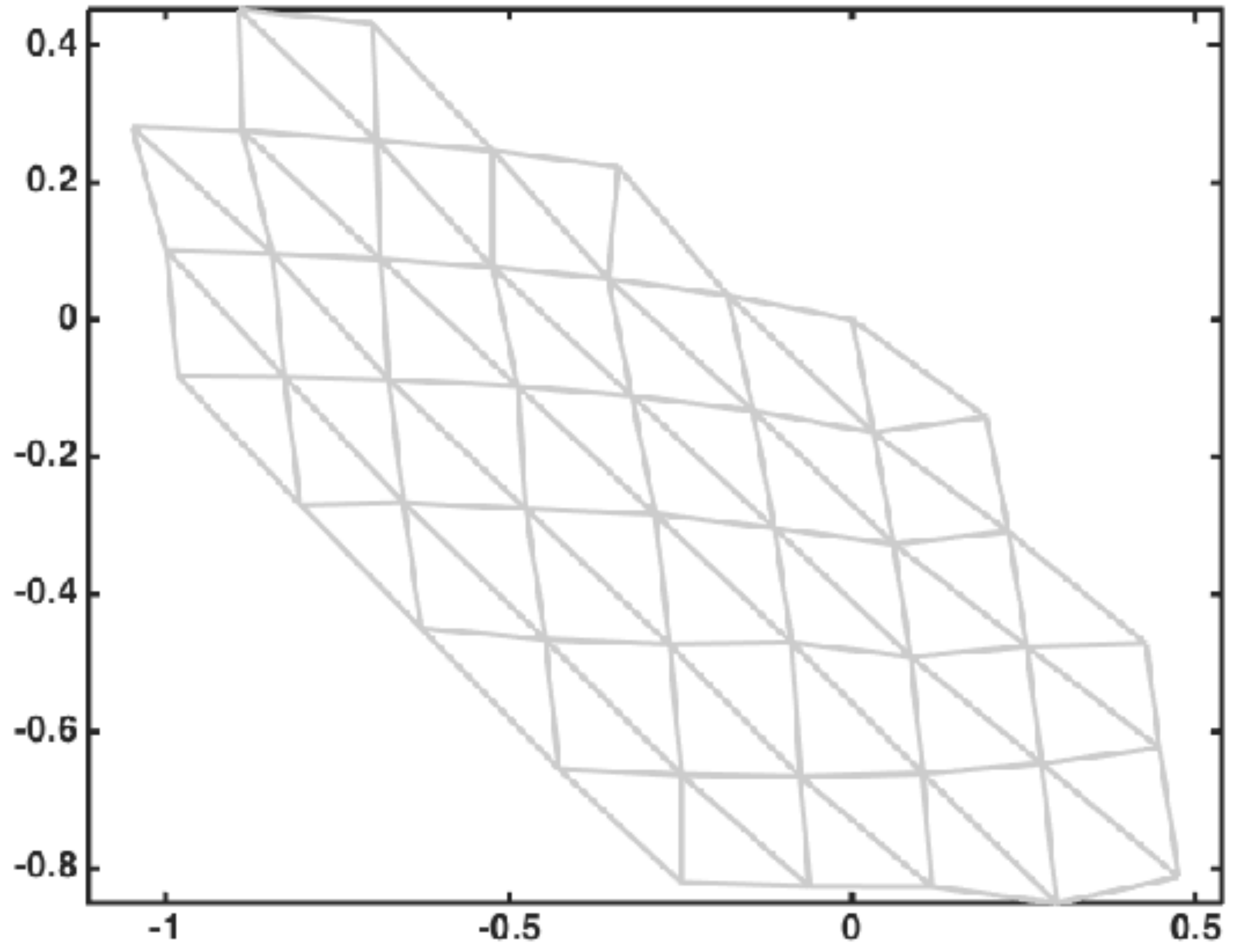
mesh




U-function



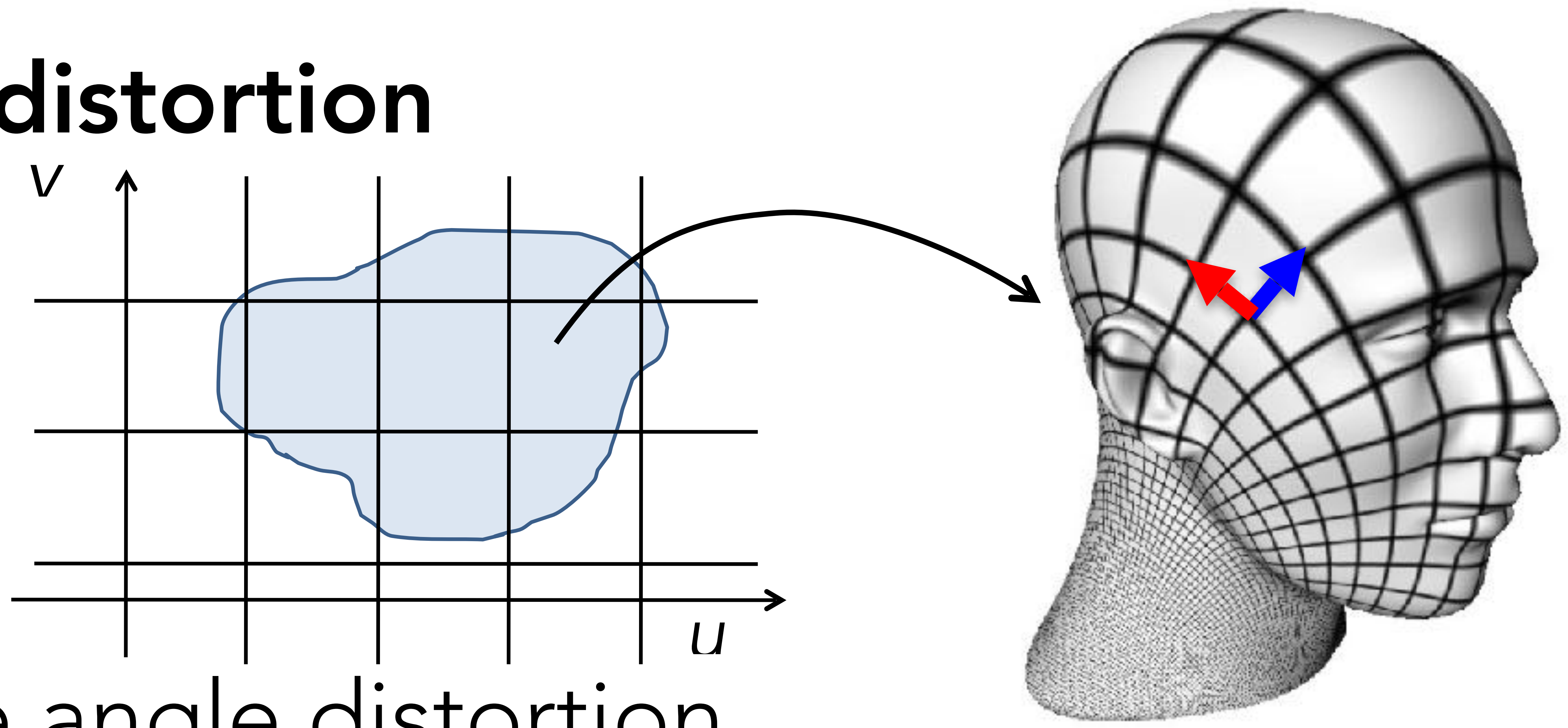
V-function




flattened mesh (UV-domain)

# Least-Squares Conformal

- **Conformal** maps have **zero angle distortion** (preserve 90° angles)




- Least-Squares Conformal: minimize angle distortion

$$\min_{u,v} \int_M \|{\color{blue}\nabla u}^{\perp} - {\color{red}\nabla v}\|^2 \,\mathrm{d}V$$

- Encourage perpendicular, equally scaled coordinate vectors



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# Least-Squares Conformal

$$\min_{u,v} \frac{1}{2}\int_M \|\nabla u^{\perp} - \nabla v\|^2 \, \mathrm{d}V$$

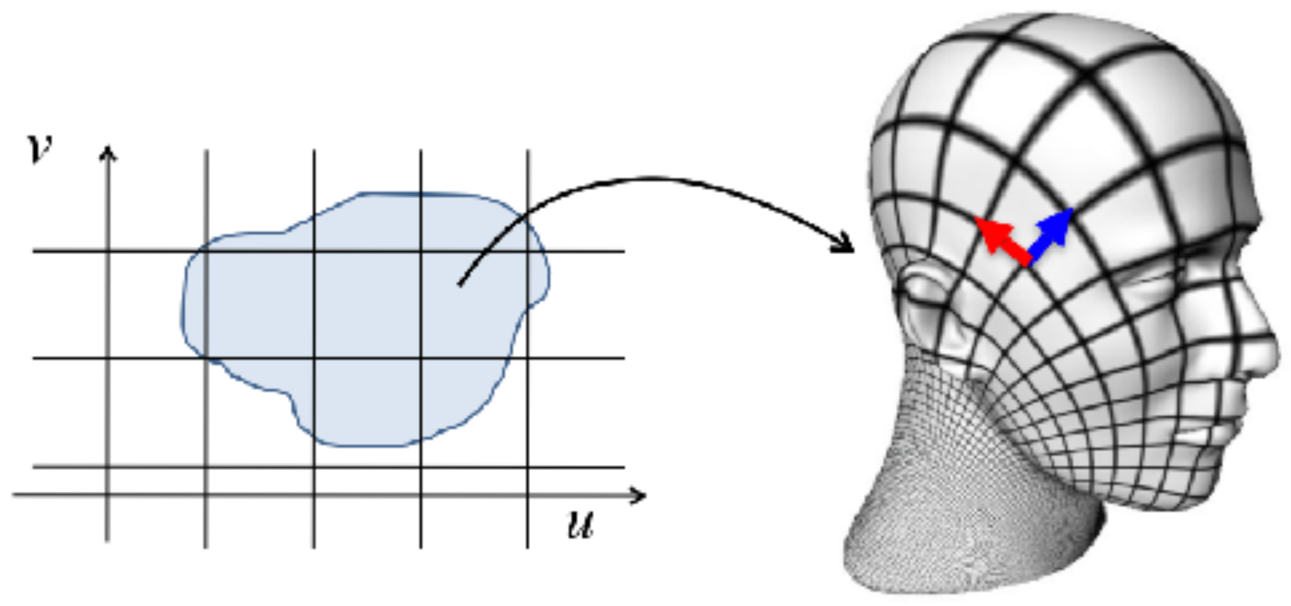
$$= \min_{u,v} \frac{1}{2}\int_M (\hat{n} \times \nabla u - \nabla v) \cdot (\hat{n} \times \nabla u - \nabla v) \, \mathrm{d}V$$

$$= \min_{u,v} \frac{1}{2}\int_M \|\hat{n} \times \nabla u\|^2 - 2(\hat{n} \times \nabla u) \cdot \nabla v + \|\nabla v\|^2 \, \mathrm{d}V$$

$$= \min_{u,v} \frac{1}{2}\int_M \|\nabla u\|^2 + \|\nabla v\|^2 - 2(\hat{n} \times \nabla u) \cdot \nabla v \, \mathrm{d}V$$

$$= \min_{u,v} \frac{1}{2}\int_M \|\nabla u\|^2 + \|\nabla v\|^2 - 2\det\left(\begin{array}{c} \nabla u^T \\ \hline \nabla v^T \\ \hline \hat{n}^T \end{array}\right) \mathrm{d}V$$

$$= \underline{\min_{u,v} \frac{1}{2}\int_M \|\nabla u\|^2 \, \mathrm{d}V + \frac{1}{2}\int_M \|\nabla v\|^2 \, \mathrm{d}V - A(u,v)}$$



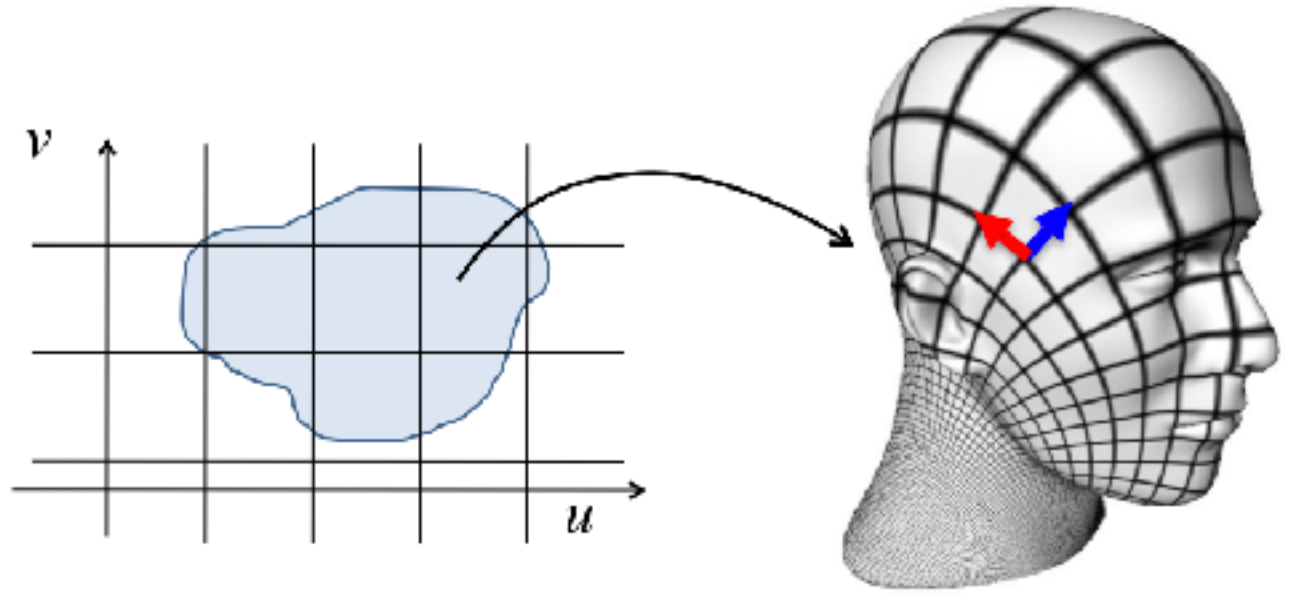

Jacobian Determinant
(Area scaling at each pt)

Area in (u,v) plane!

# LSCM ==> Harmonic

- We showed LSCM decomposes into:

$$\frac{1}{2}\int_M \|\nabla u^\perp - \nabla v\|^2 \,\mathrm{d}V = \underbrace{\frac{1}{2}\int_M \|\nabla u\|^2 \,\mathrm{d}V + \frac{1}{2}\int_M \|\nabla v\|^2 \,\mathrm{d}V}_{\text{Smoothness of } u \text{ and } v \text{ ("Dirichlet Energy")}} - \underbrace{A(u, v)}_{\text{Area in plane}}$$



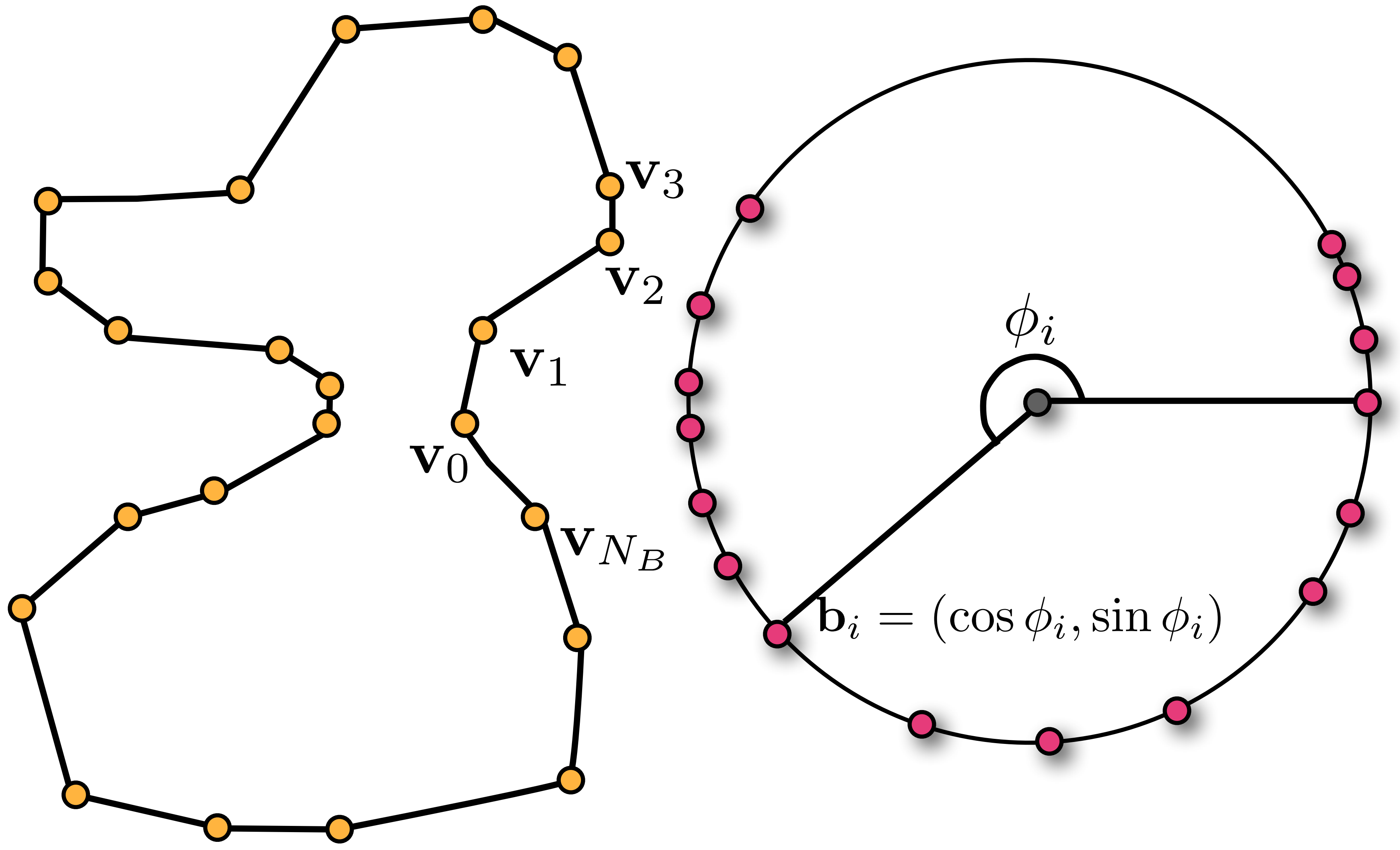

Smoothness of u and v
("Dirichlet Energy")

Area in
plane

- If we fix the **flattened boundary** (area), LSCM is equivalent to designing **smooth scalar fields**

# Harmonic Parametrization

- Map mesh boundary to a convex polygon in plane

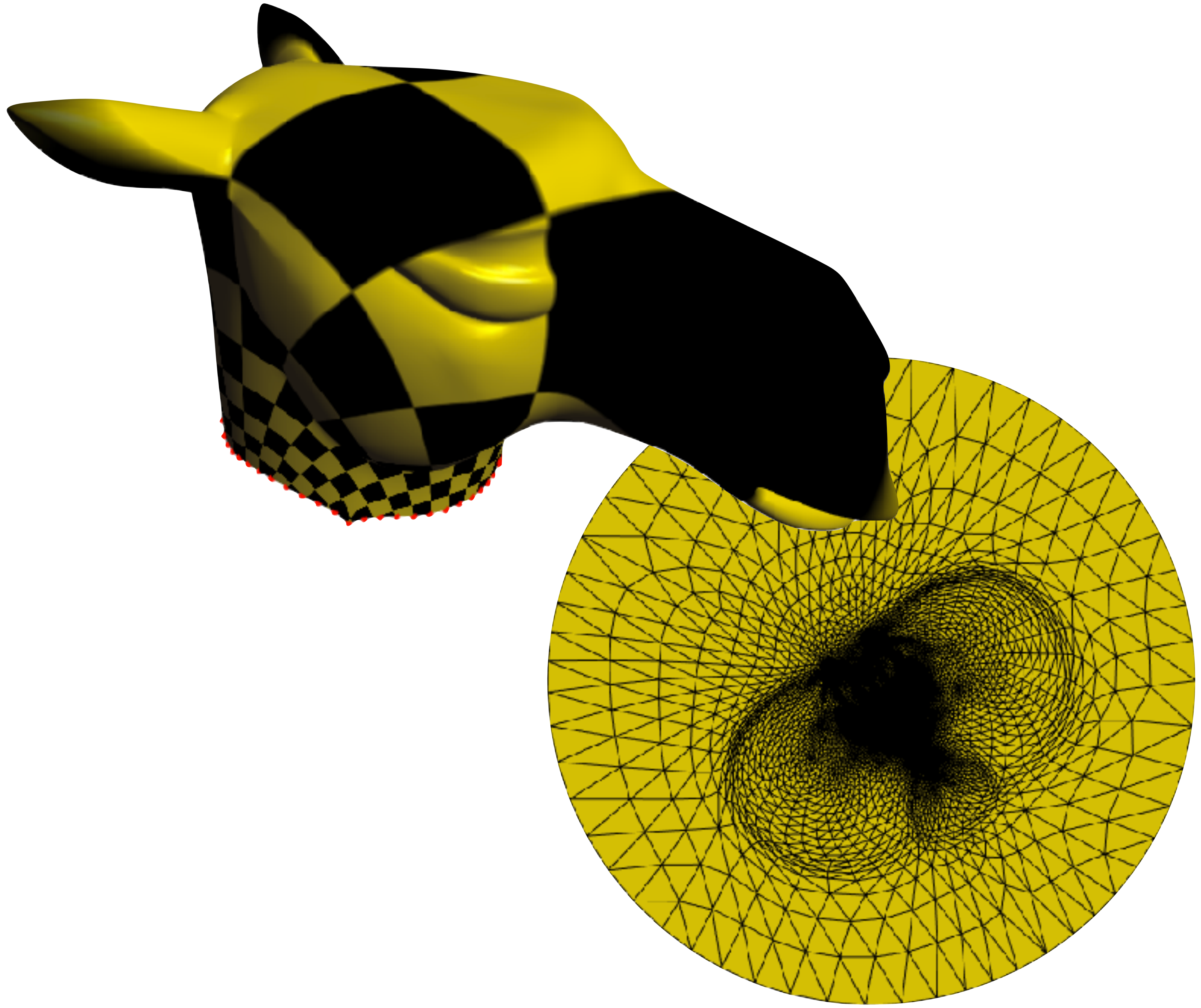


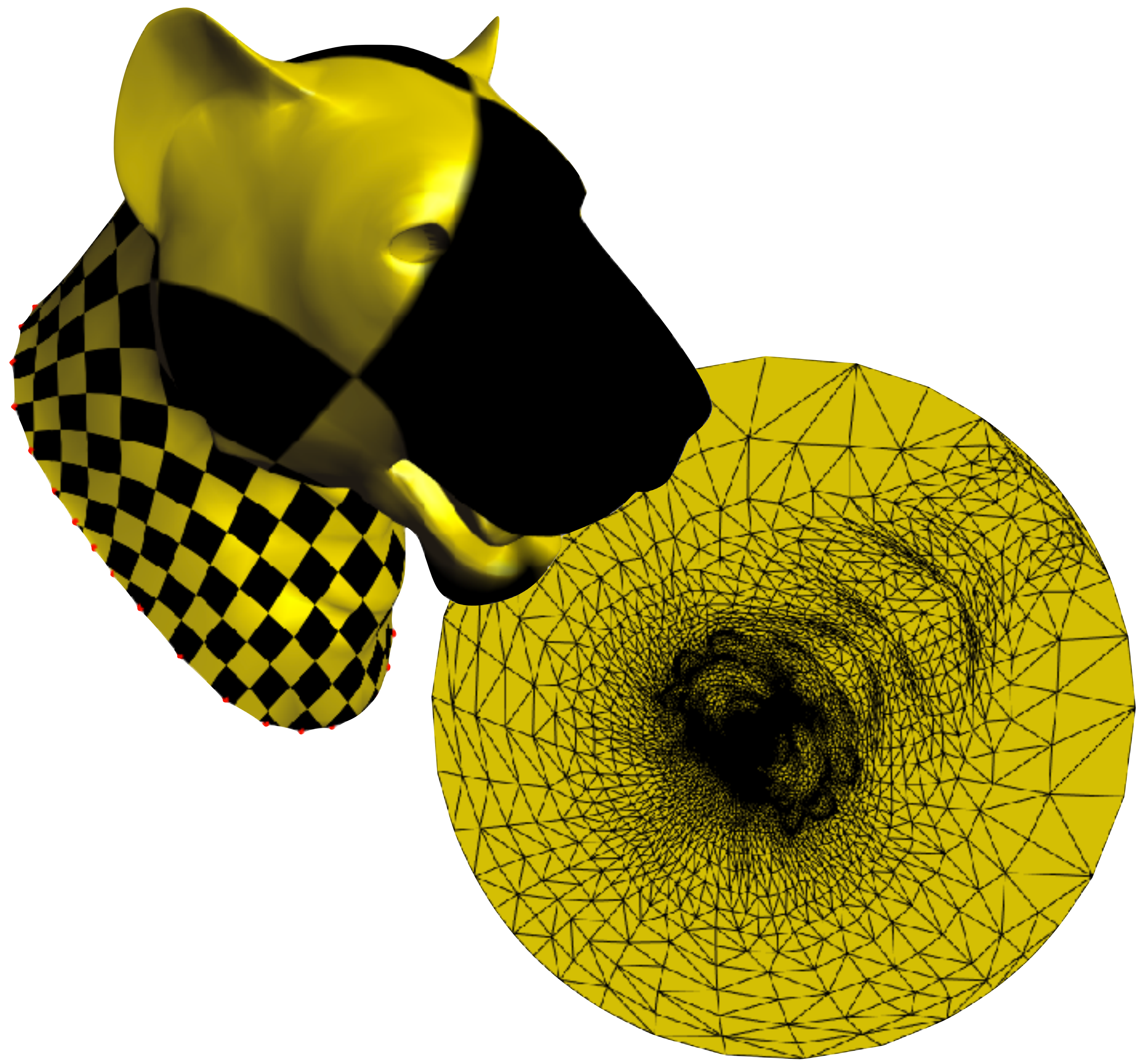

- Solve for smoothly interpolating (u, v):

$$\min_{u=u_b \text{ on } \partial M} \frac{1}{2}\int_M \|\nabla u\|^2 \, \mathrm{d}V \quad \Longrightarrow \quad \begin{cases} \Delta u = 0 & \text{in } M \\ u = u_b & \text{on } \partial M \end{cases}$$

Just solve Laplace equation with fixed boundary values!

# Harmonic Parametrization

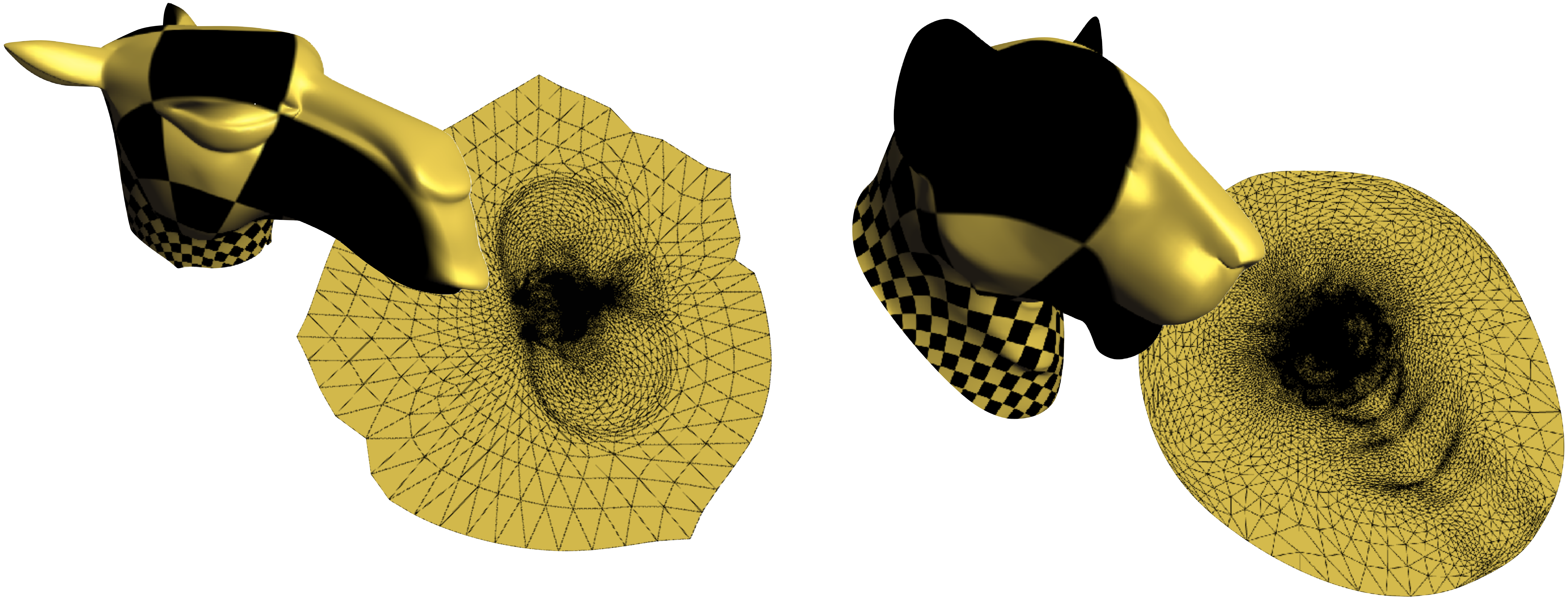




Already implemented in libigl!

See tutorial Chapter 4

# LSCM Parametrization



Already implemented in libigl!

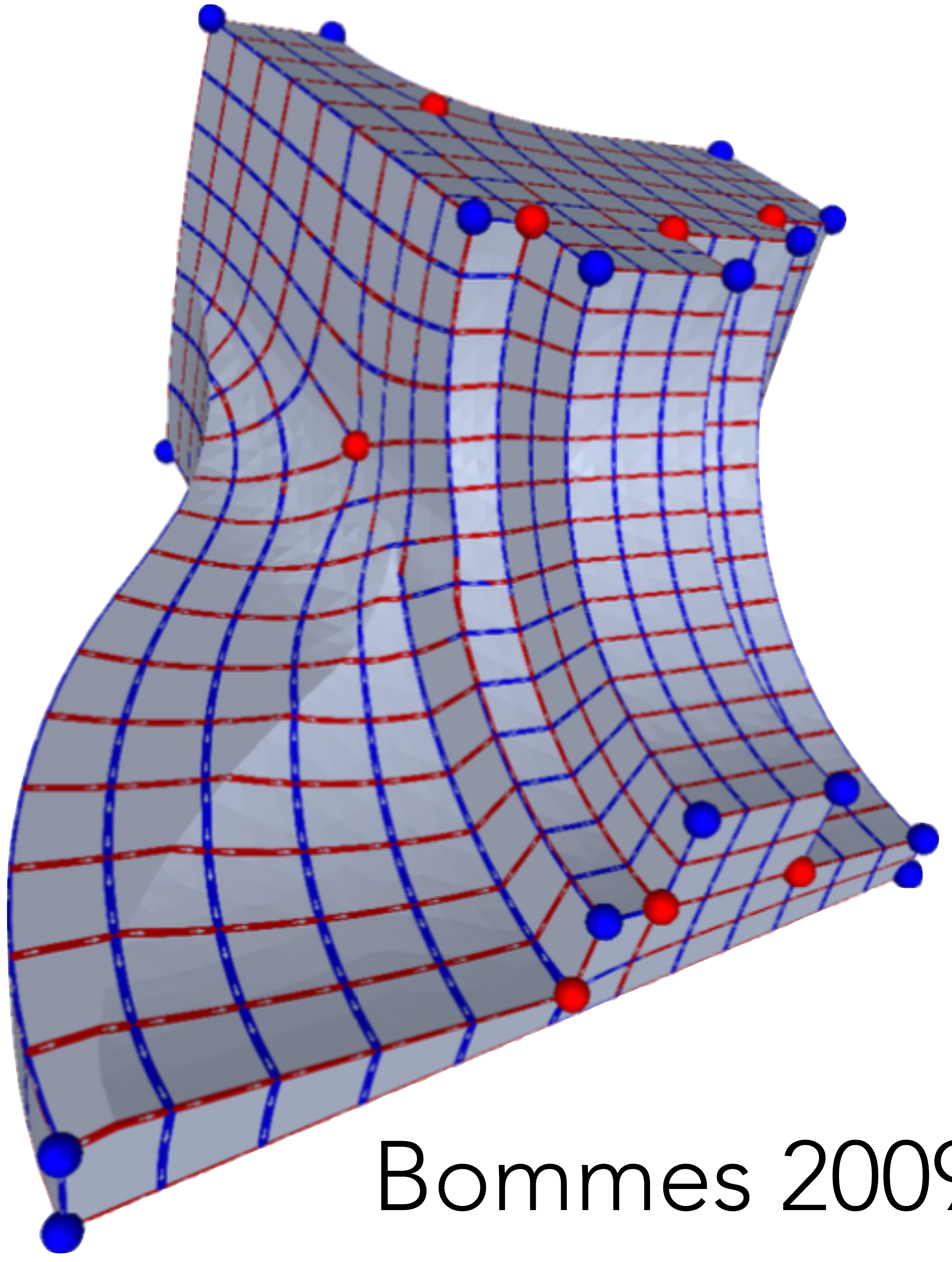
See tutorial Chapter 4

# Field-Guided Parametrization

- Often we want coordinate directions to **align** with mesh features.

- We can do this by **specifying vector fields** and **fitting (u, v) gradients** (coordinate directions) to them.

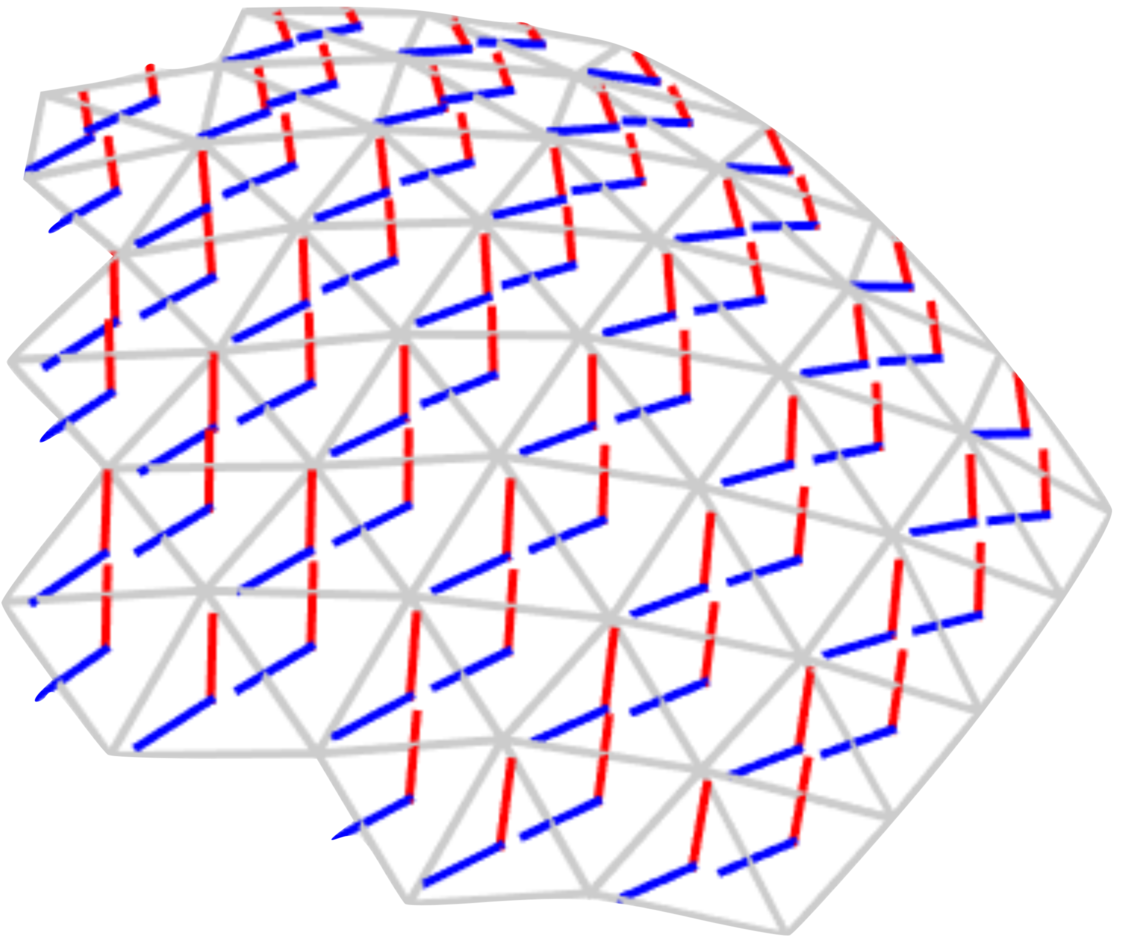
- Orthonormal vectors => minimize **angle and area distortion**



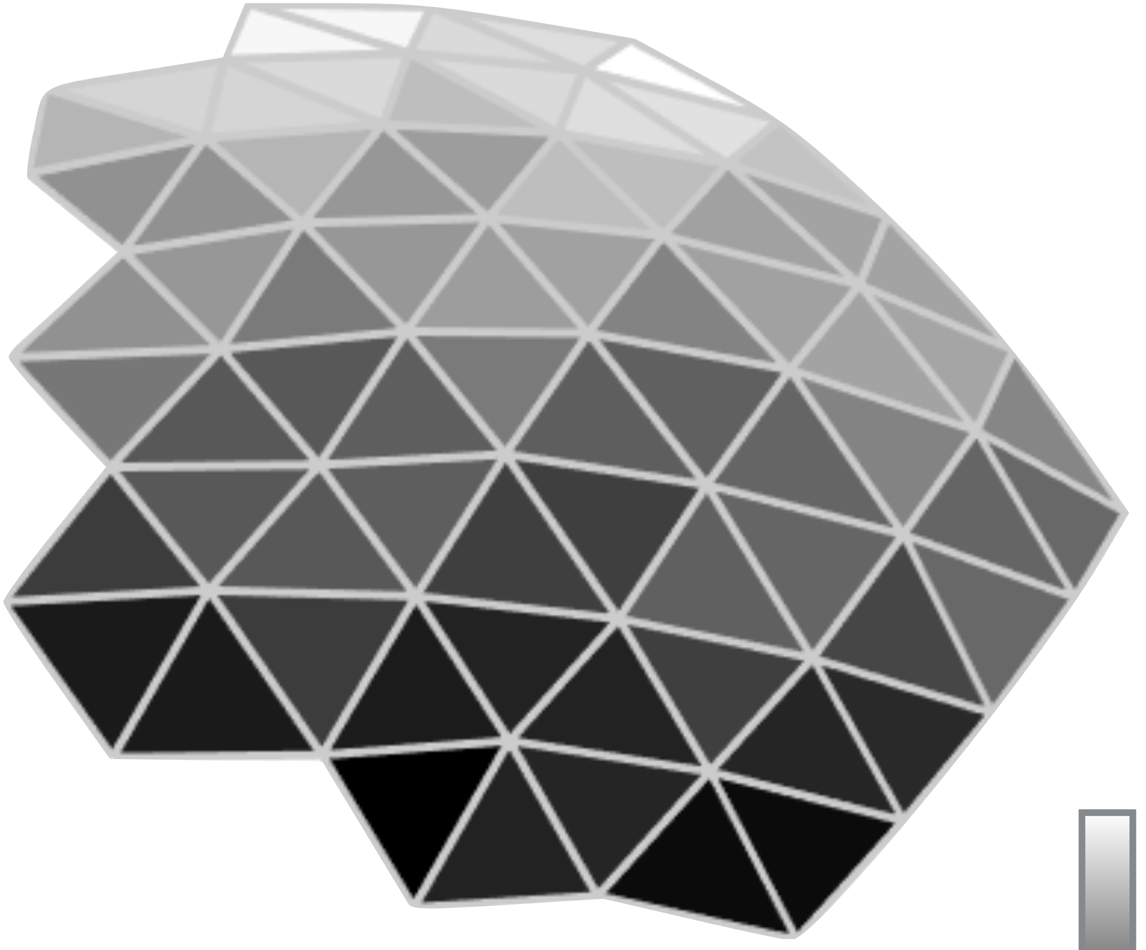
Bommes 2009

# Field-Guided Parametrization

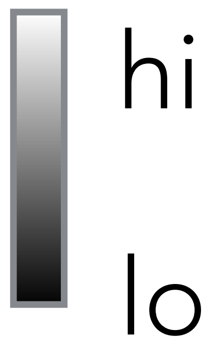
- We'll consider piecewise linear (per-vertex) scalar fields, so gradients and vector fields are constant per triangle:

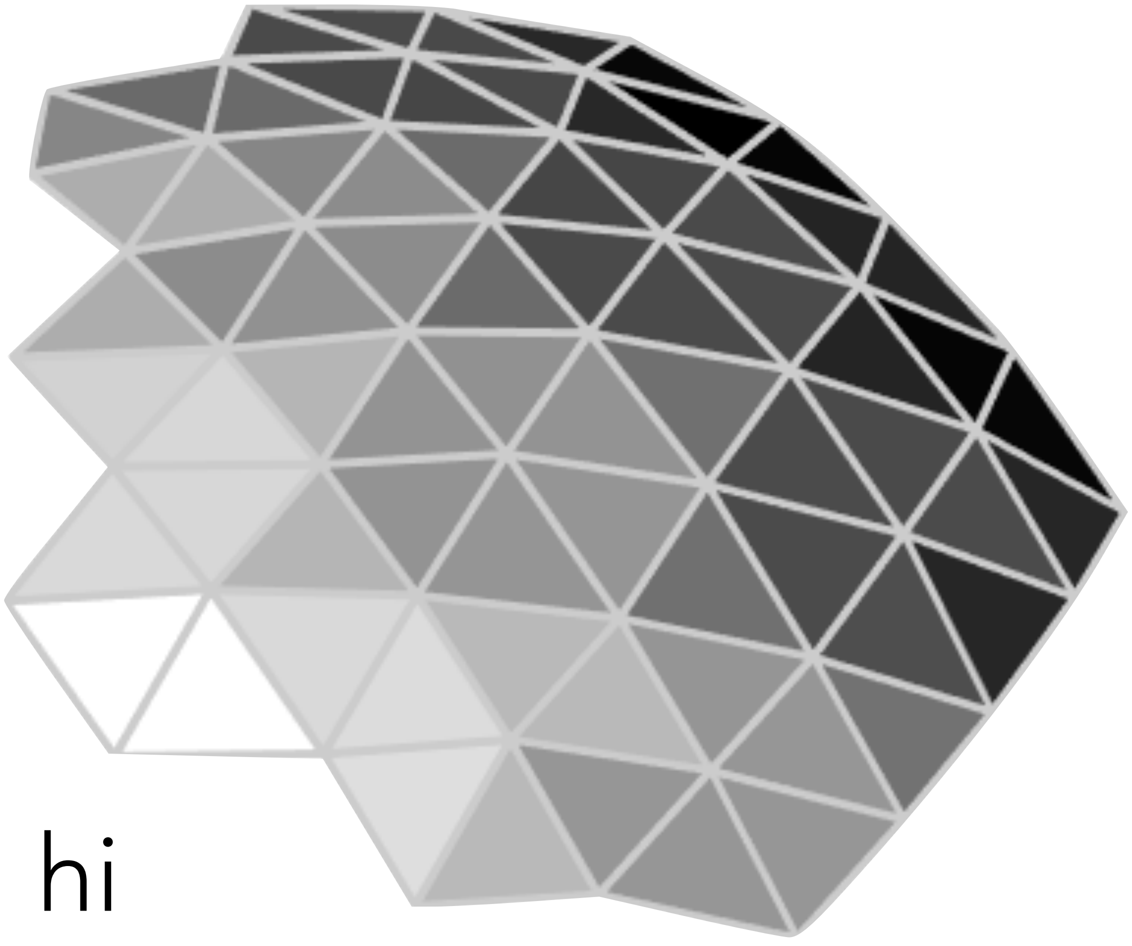


mesh

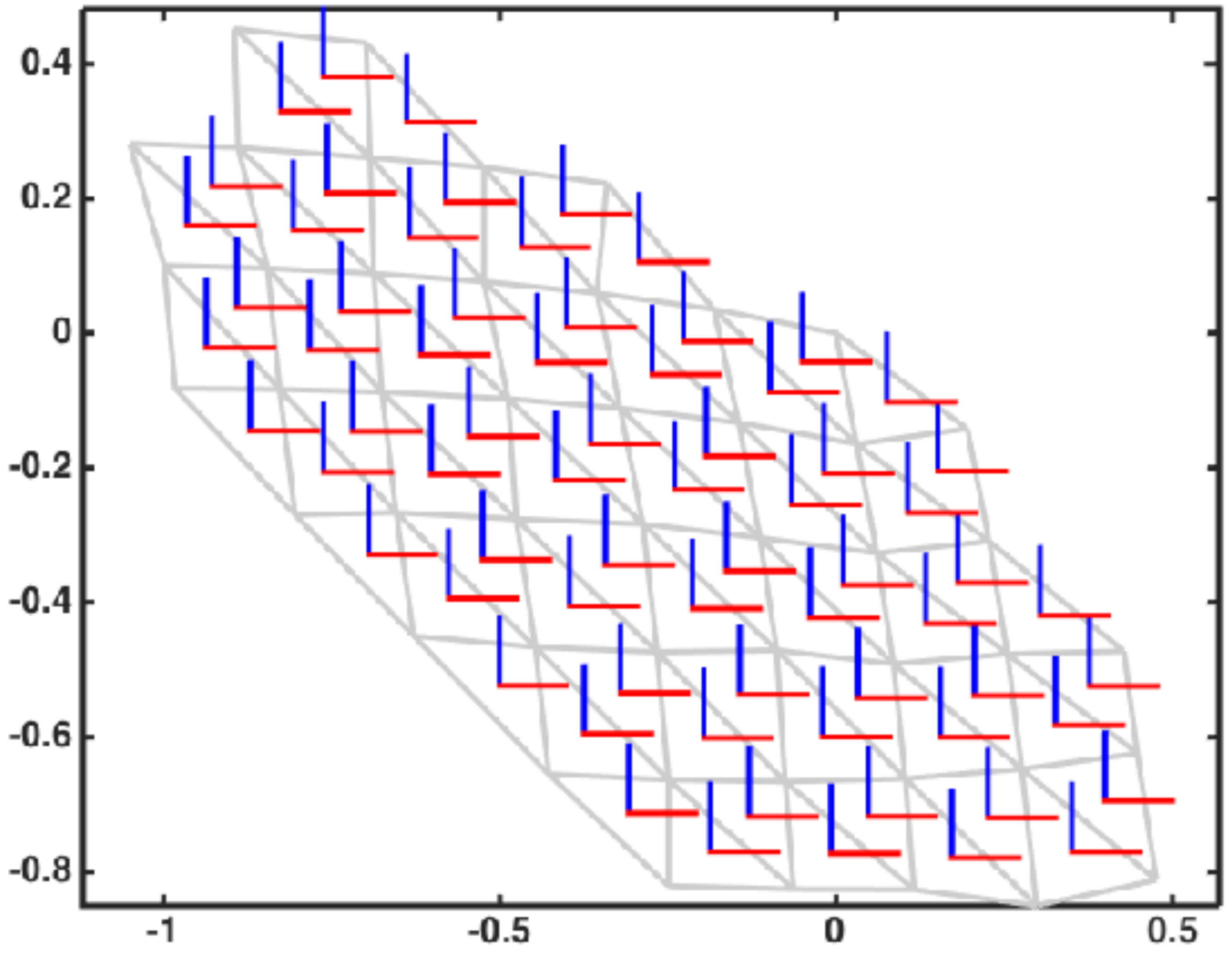


U-function



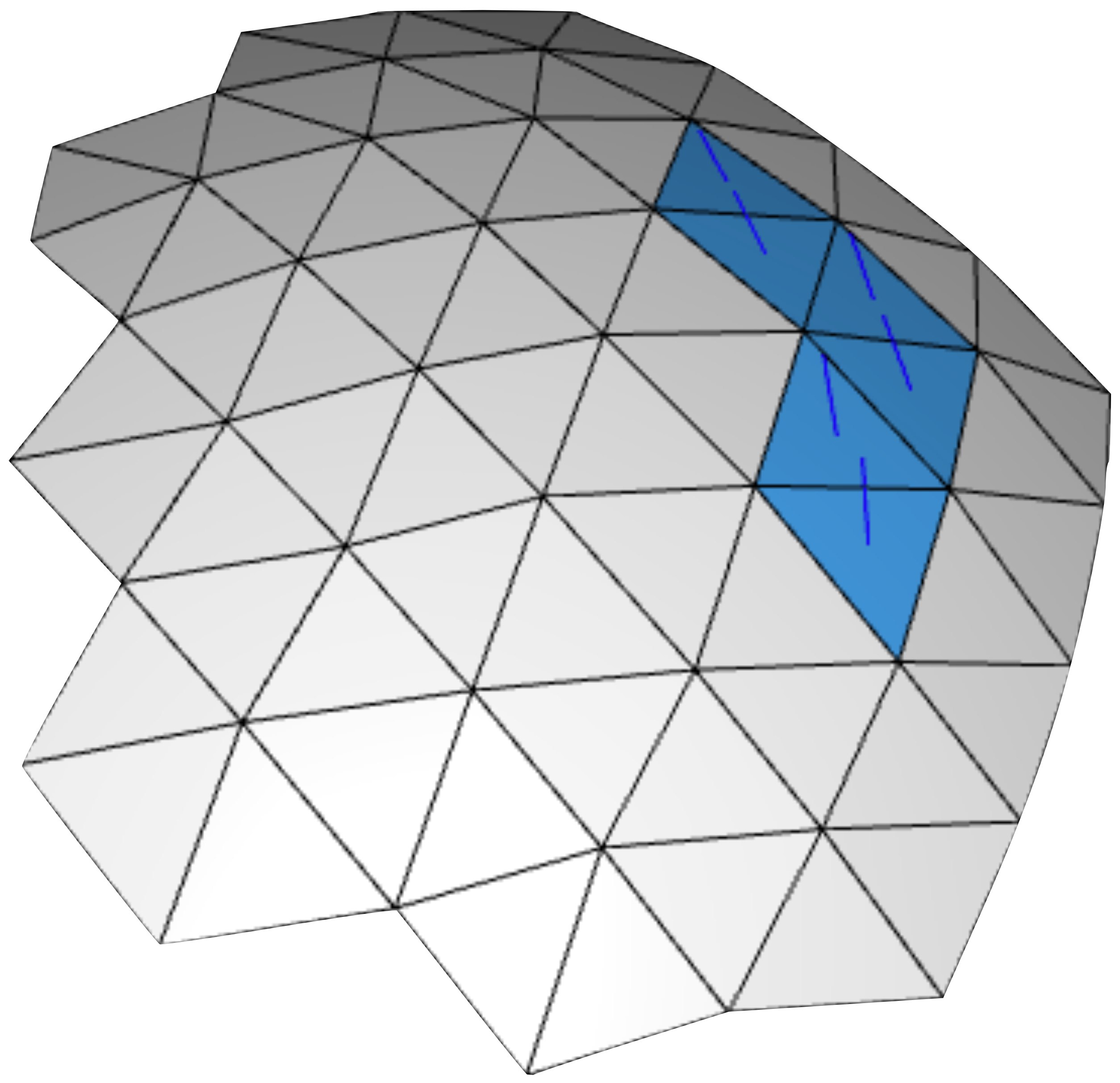
V-function



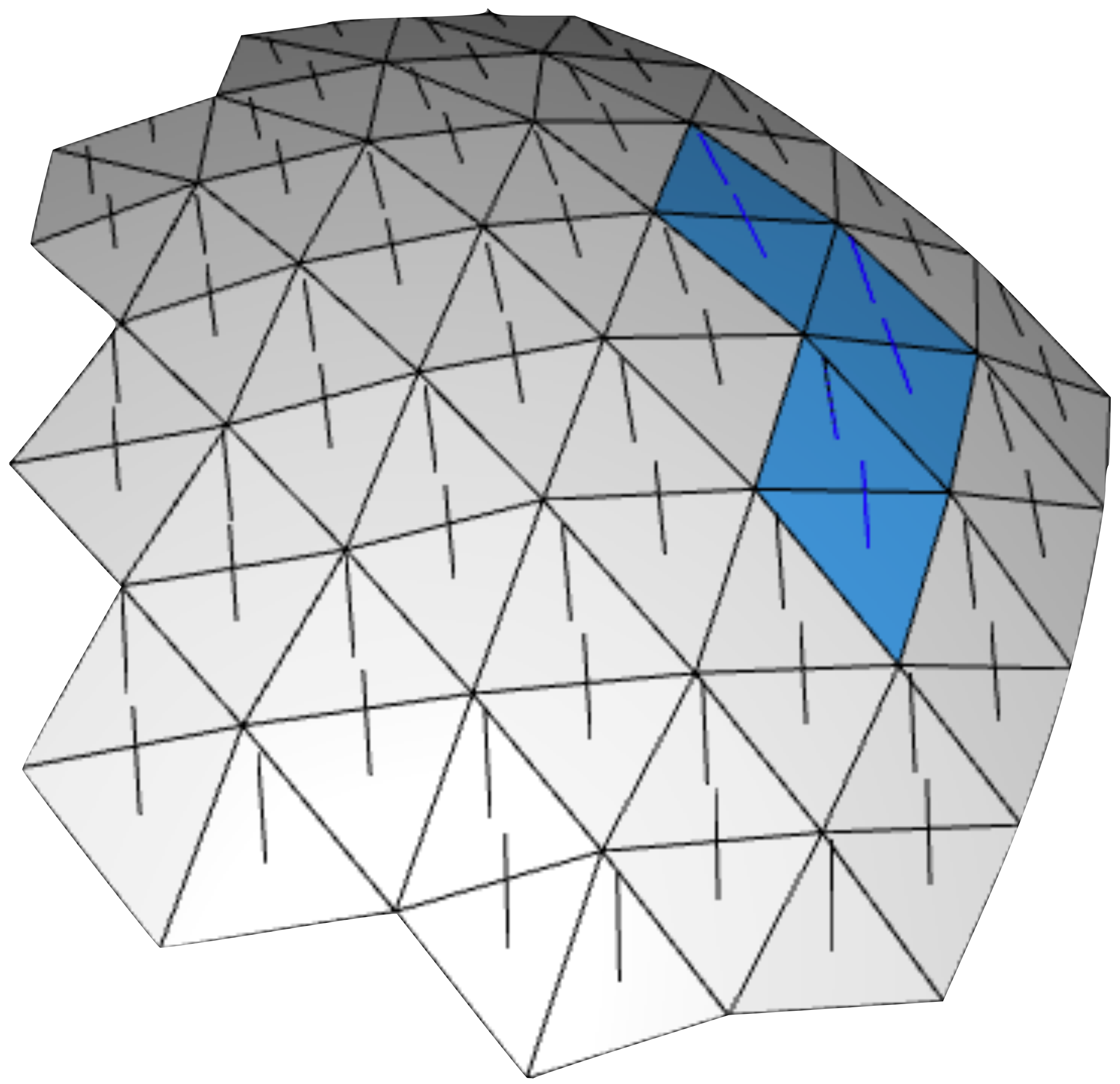

flattened mesh (UV-domain)

# Vector-Guided Scalar Fields

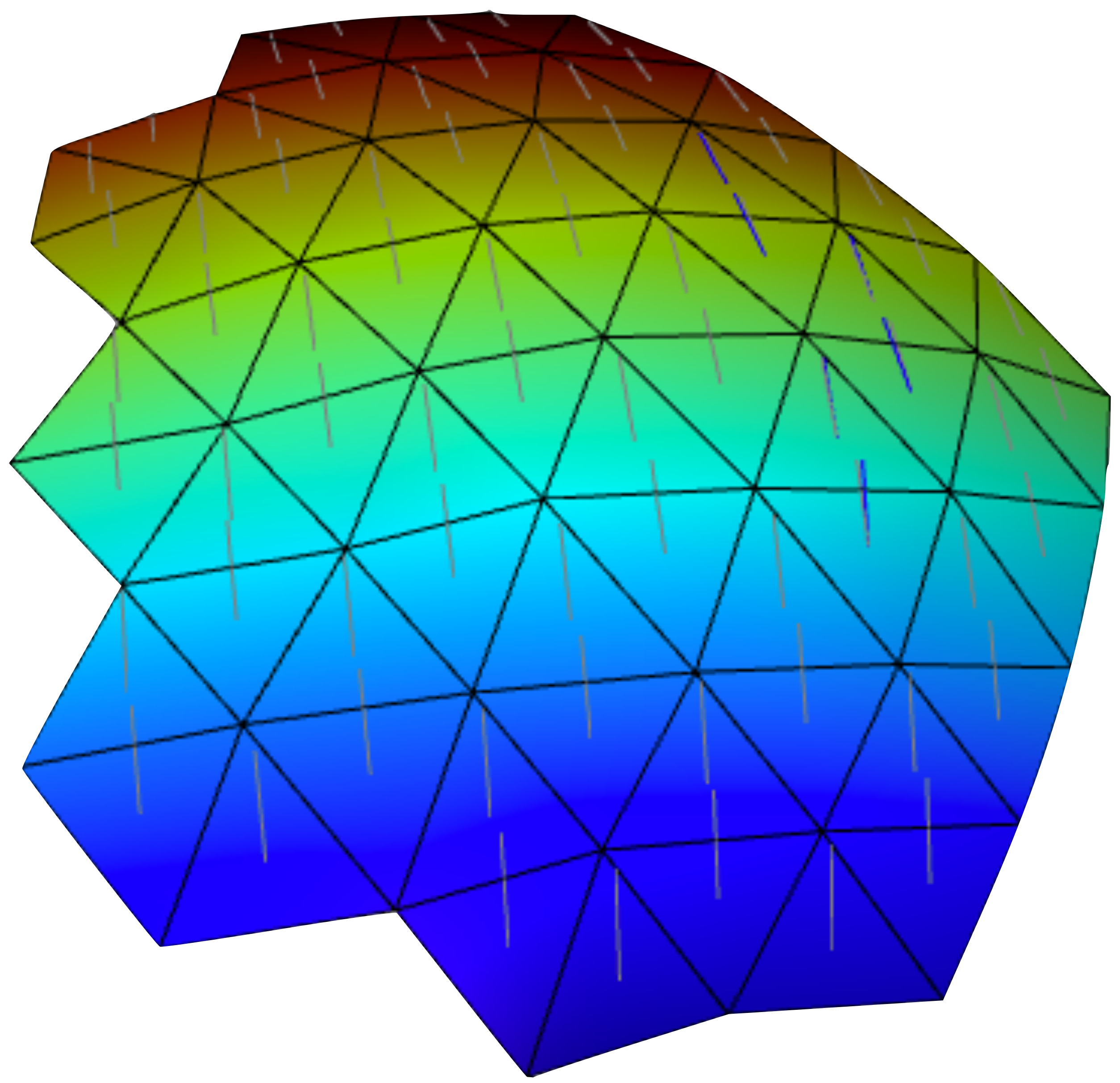
- Main HW4 task: design smooth vector fields and compute a scalar field aligned to them.



vector constraints

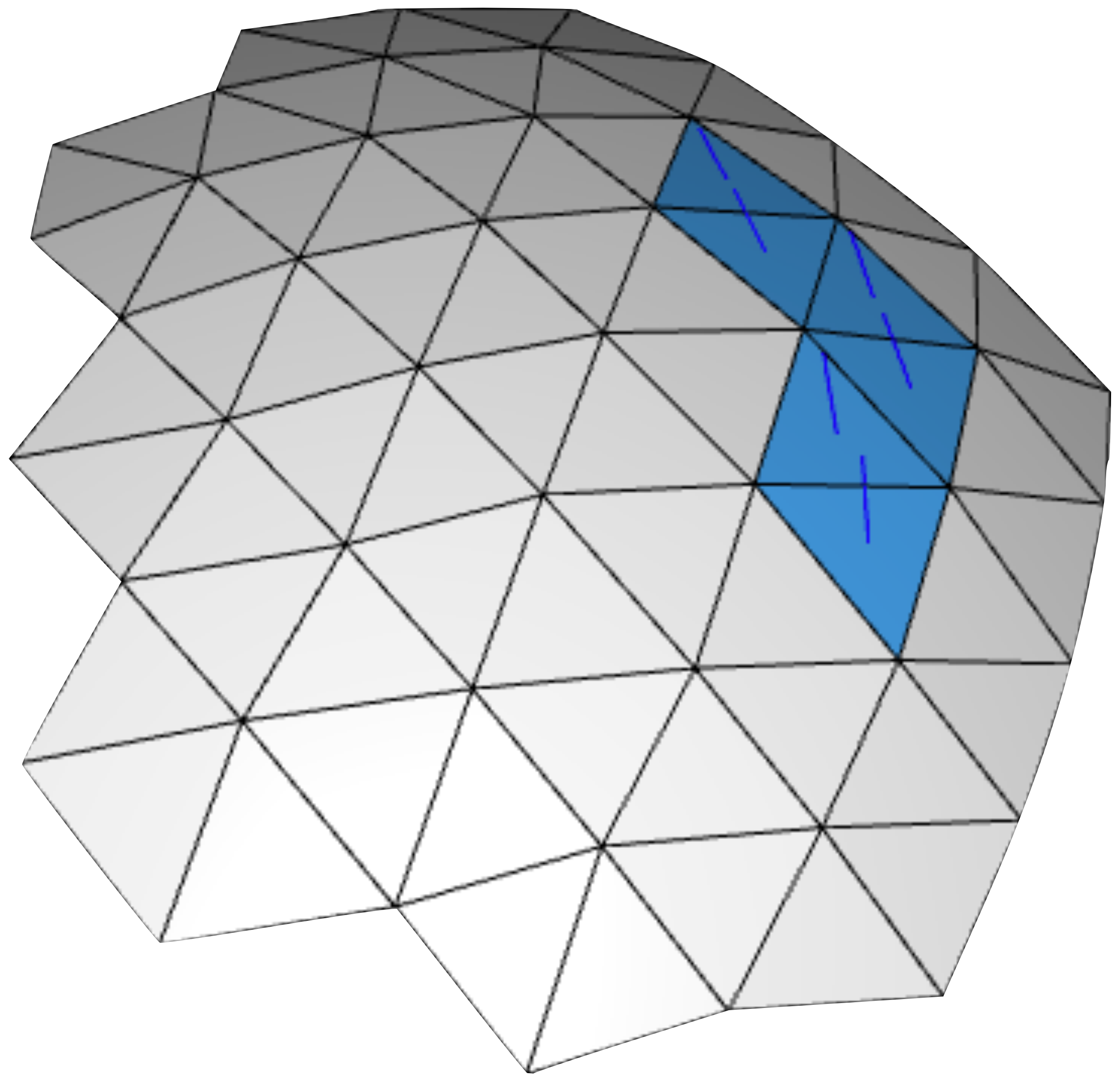


smooth interpolated field

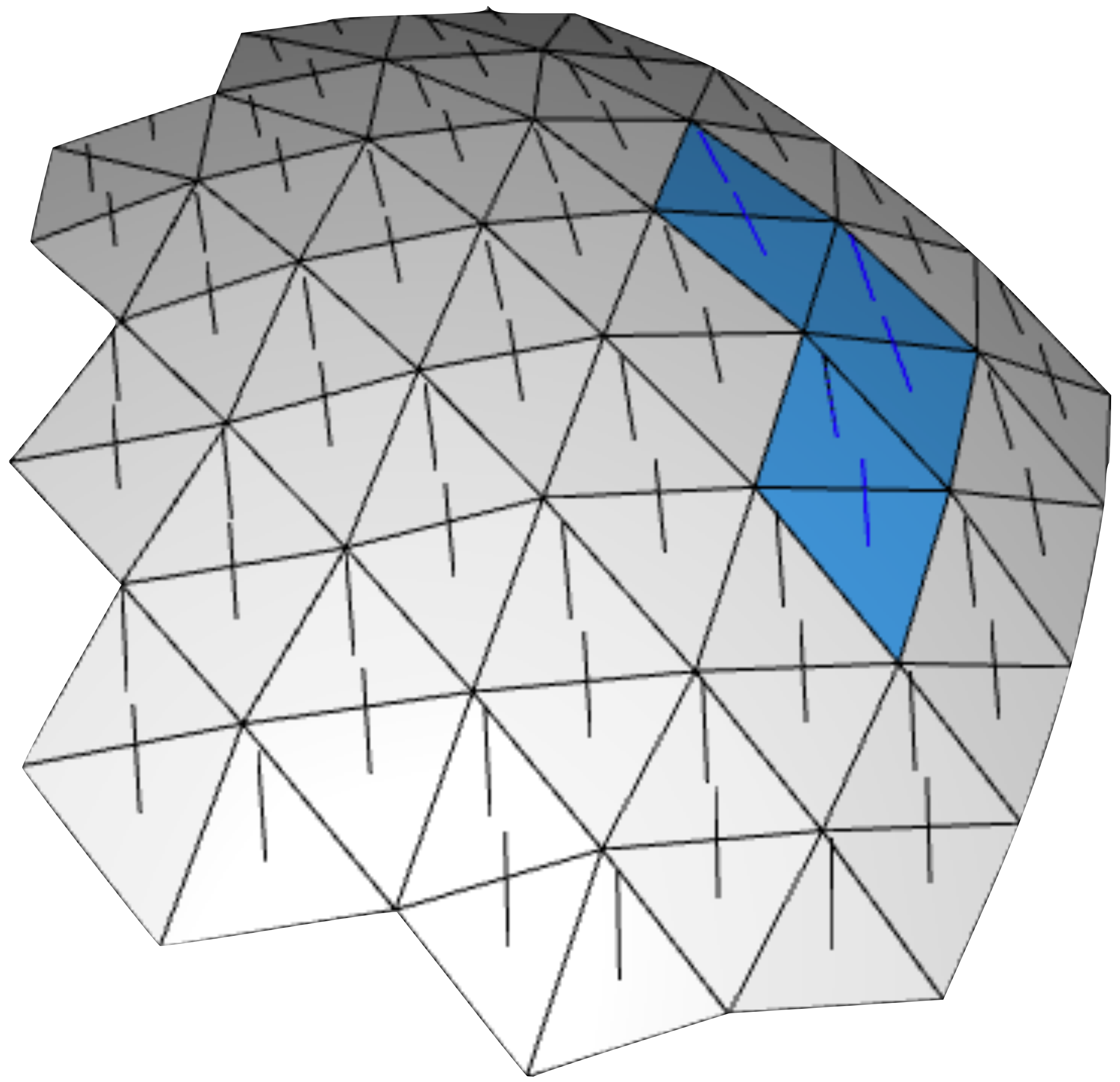


reconstructed scalar function and gradient

# Step 1: Vector Field Design



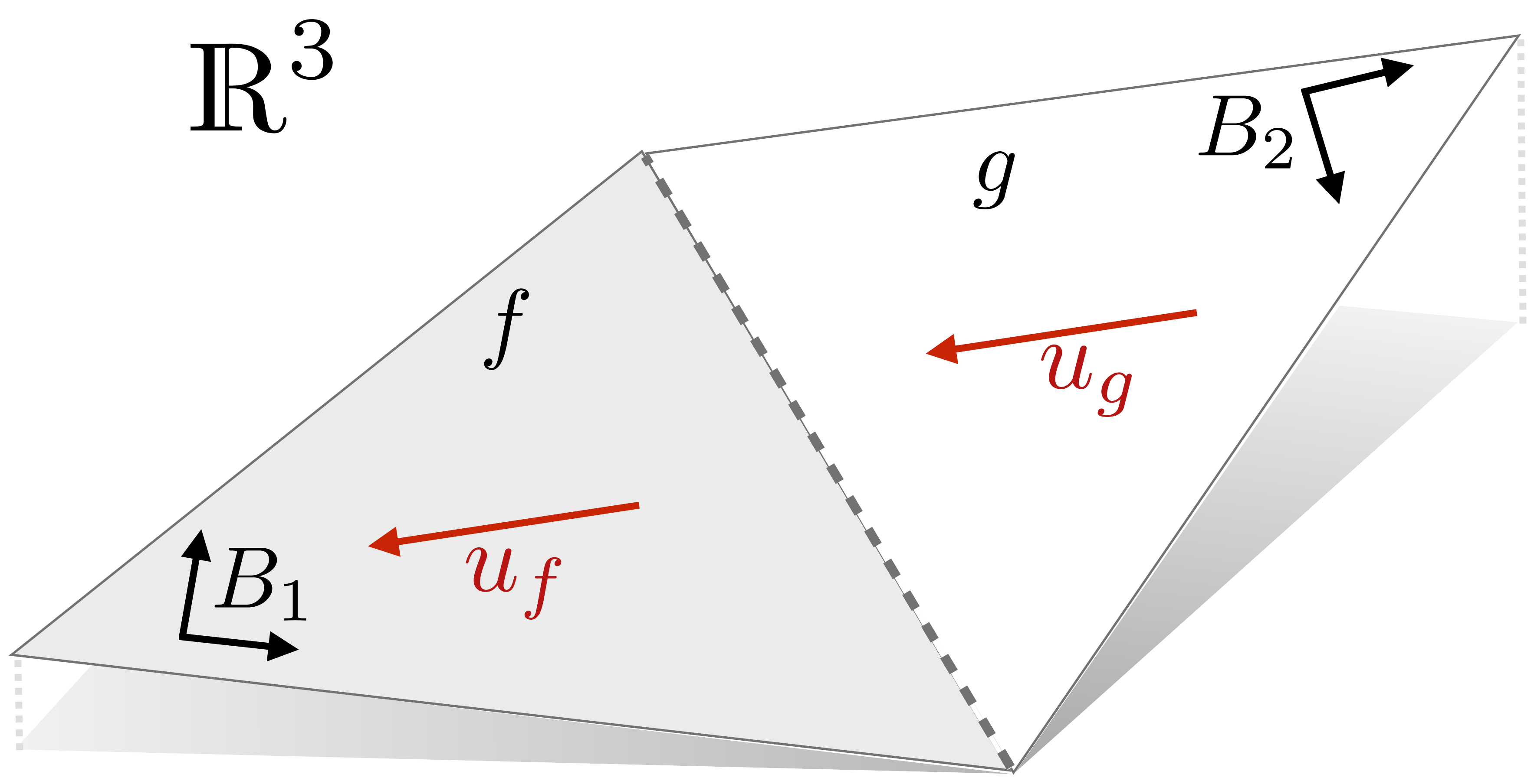
Paint constraint vectors
on some faces

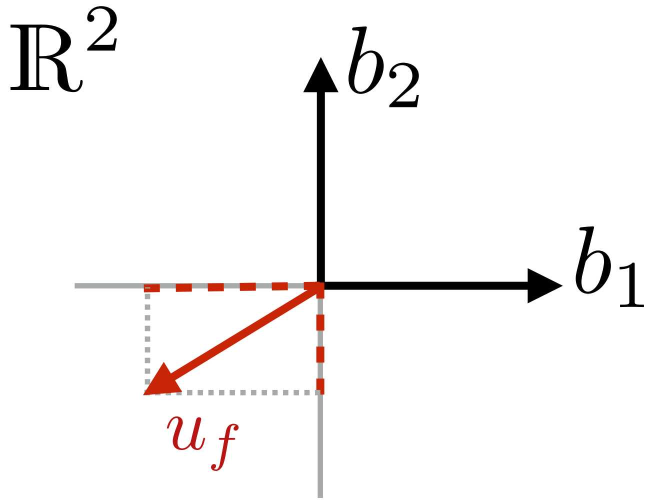


Smoothly interpolated
field

# Vector Field Representation

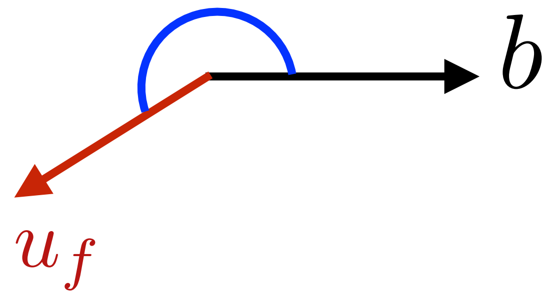
- Tangent vector per triangle:
  represent in an arbitrary local basis for each triangle:




$$u_f = (u_f^x, u_f^y) \in \mathbb{R}^2$$

$\mathbb{C}$




Real Axis

$$u_f = u_f^x + i u_f^y \in \mathbb{C}$$

- (Optional) notation simplification:
  Vector = complex number
  (Tangent plane = complex plane)



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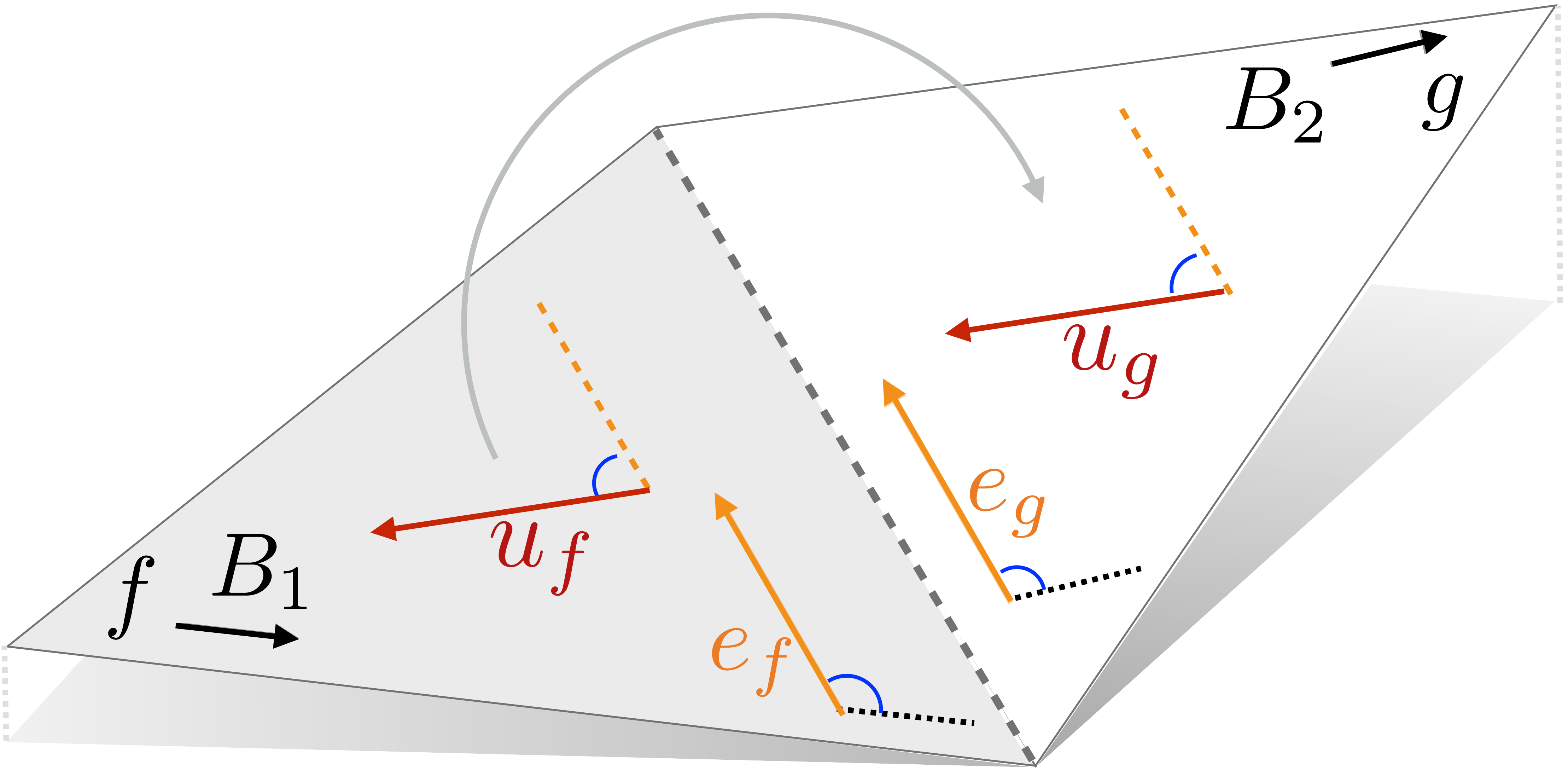
# Vector Field Smoothness

- Try to keep vectors constant
- Vectors lie in different tangent planes! Must transport to compare.
- Idea: unfold triangle pair along shared edge, compare vectors
- Equivalent: represent vectors in orthonormal basis determined by shared edge.
  (Rotate so shared edge is real axis)




Vectors in common basis:

$$\tilde{u}_f = u_f \bar{e}_f$$
$$\tilde{u}_g = u_g \bar{e}_g$$
$$|\tilde{u}_f - \tilde{u}_g|^2$$

Since for unit complex numbers:

$$e_f^{-1} = \overline{e_f}$$



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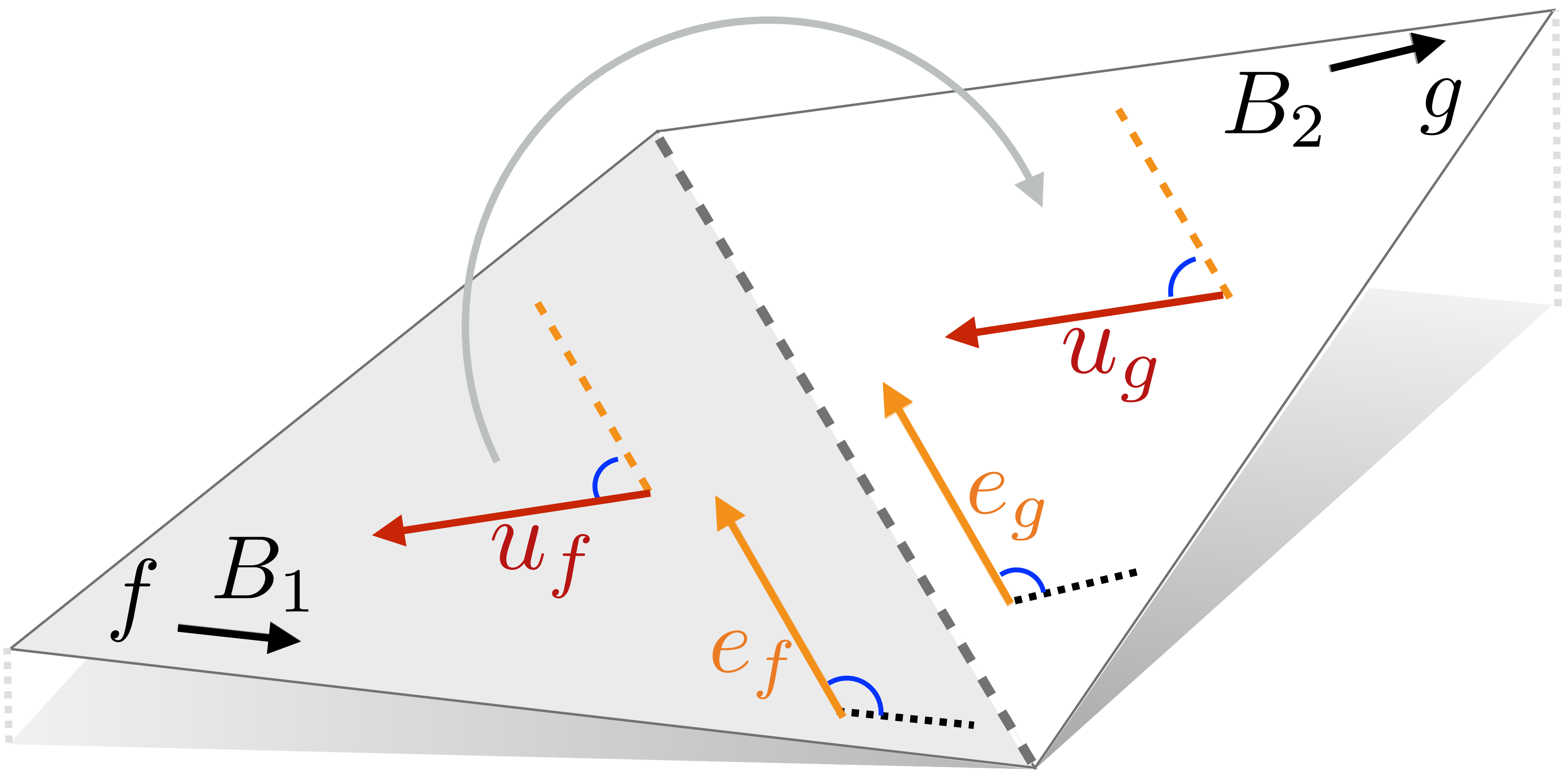
# Vector Field Smoothness




- Minimize non-smoothness across all edges:

$$\sum_{\text{edge }(f,g)} \left|u_f\overline{e_f} - u_g\overline{e_g}\right|^2 = \sum_{\text{edge }(f,g)} \begin{bmatrix}\overline{u_f} & \overline{u_g}\end{bmatrix} Q_{fg} \begin{bmatrix}u_f \\ u_g\end{bmatrix}$$

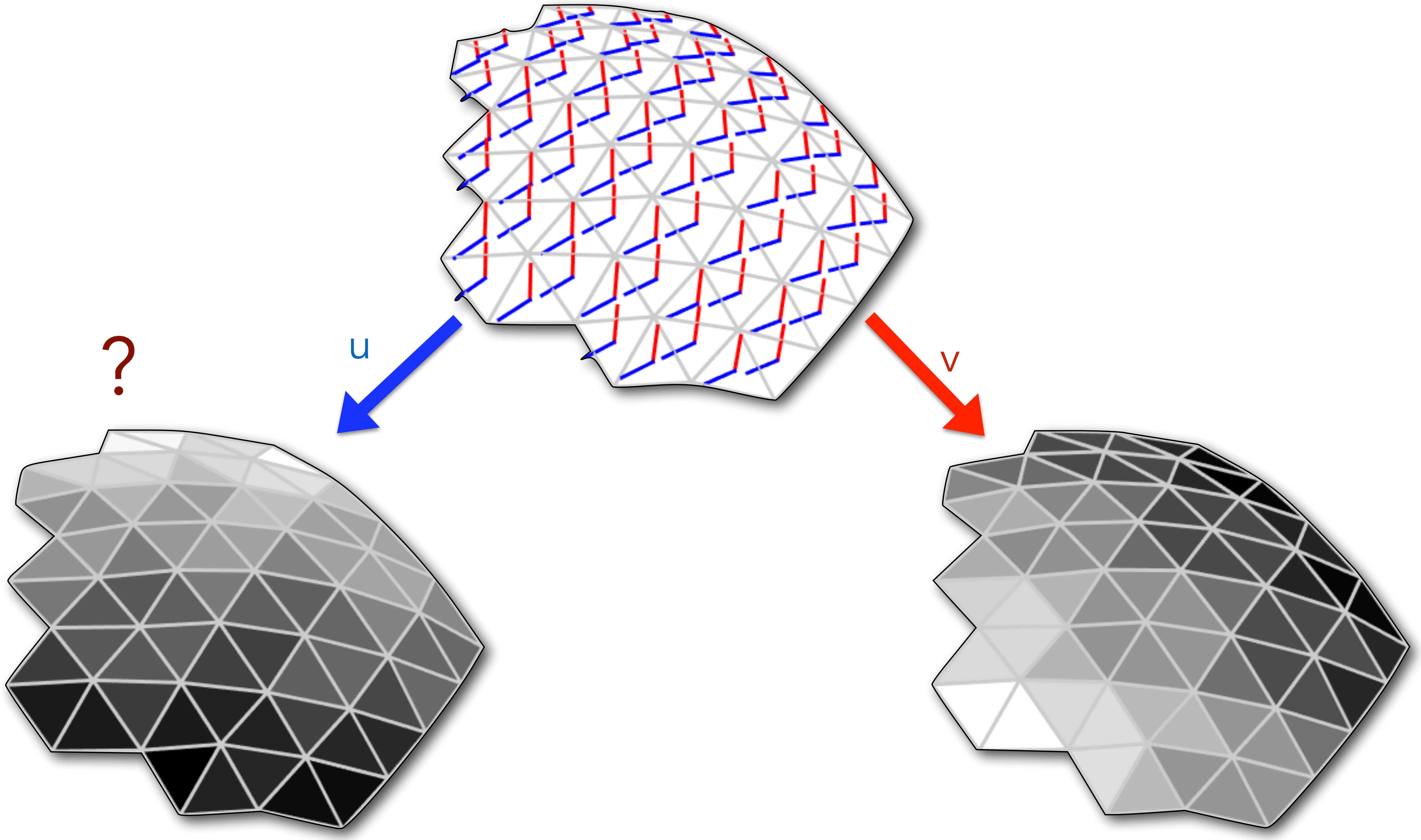
- Can be re-written as one large complex quadratic form:

$$\min_{\mathbf{u}|_{cf}=\vec{c}} \mathbf{u}^* Q \mathbf{u} \quad \Longleftrightarrow \quad \frac{\partial}{\partial \overline{\mathbf{u}}} \mathbf{u}^* Q \mathbf{u} = Q\mathbf{u} = 0$$

Complex equiv to gradient

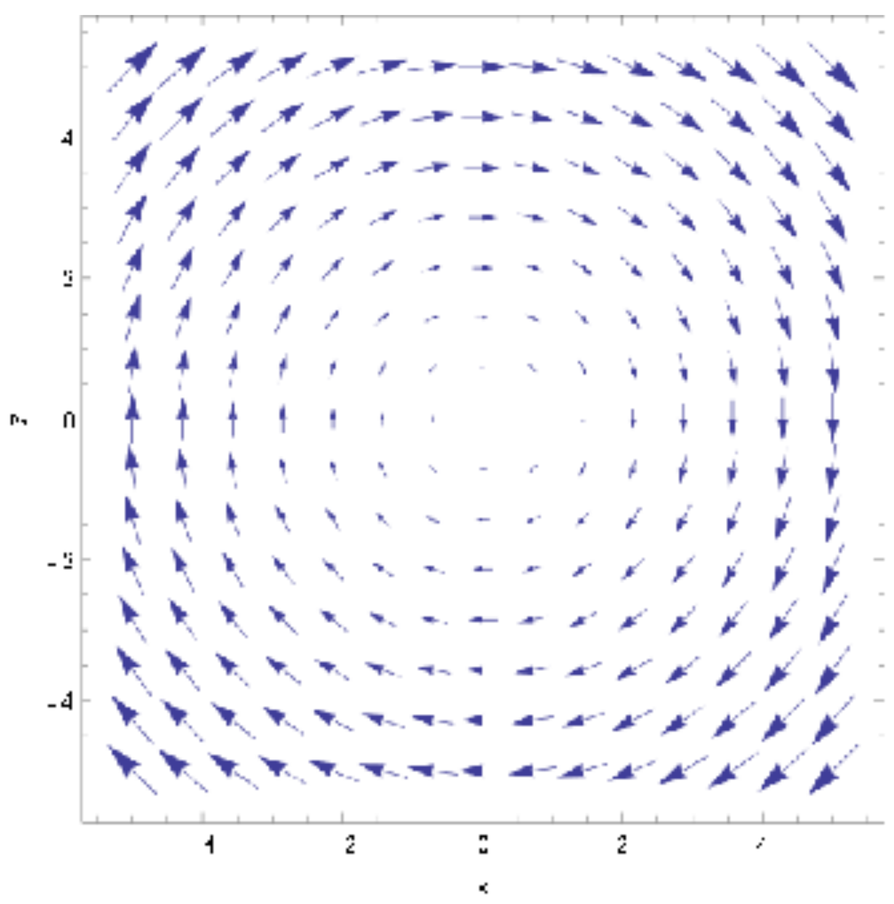
# Step 2: Scalar Field Design

- Find a scalar field whose gradients match the vector field:

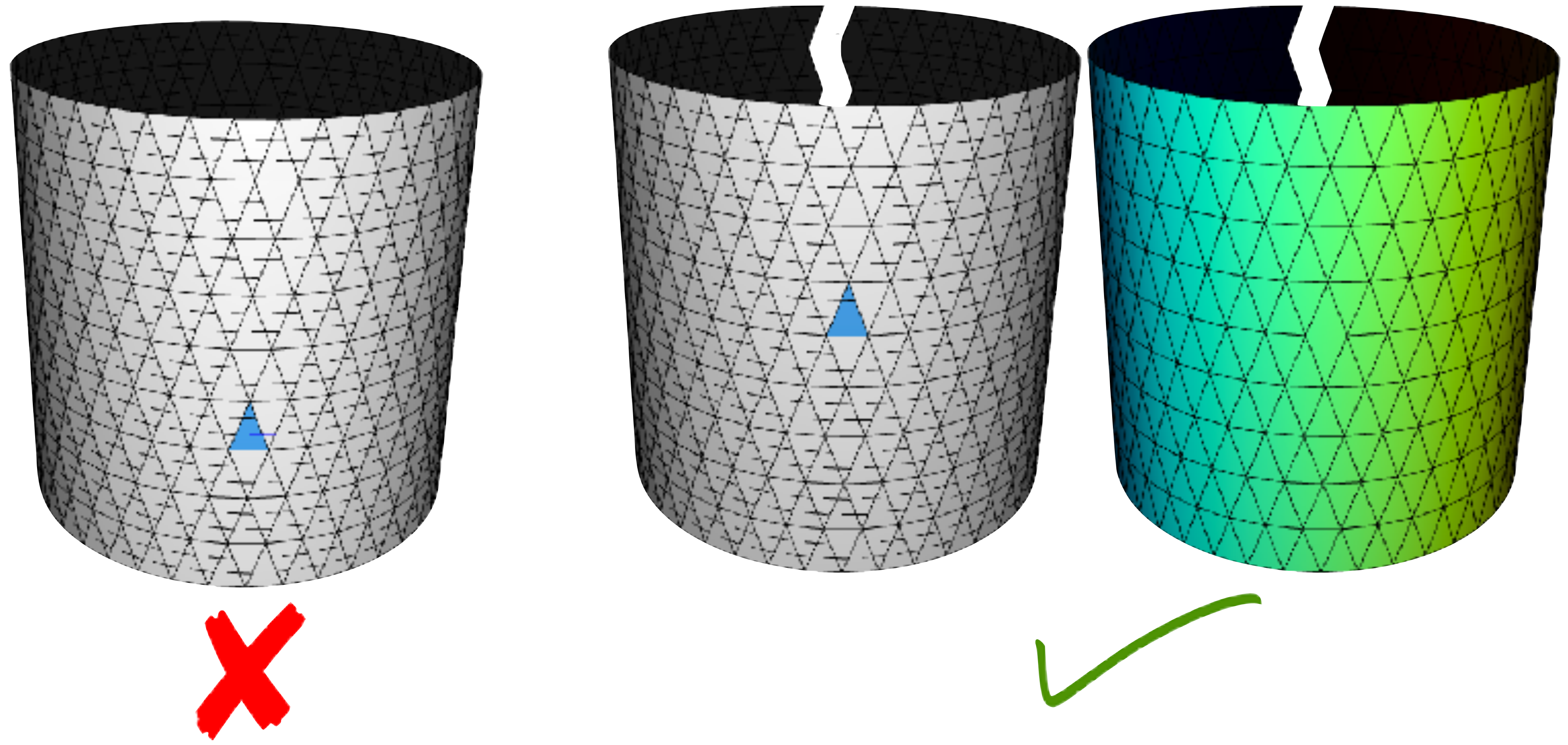


# Integrating the Vector Field

- Why not just assign zero to some vertex and "integrate" the vector field outward to get all other vertex values?
- Problem: most vector fields are not actually integrable (i.e. not the gradient of a scalar field)
- This can happen for local or global reasons:



Local: nonzero curl



Global: non-simply-connected domains



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# Integrating the Vector Field

- The fix: integrate in the "least-squares sense"

$$\min \sum_{\text{face } t} w_t \|\mathbf{g}_t - \mathbf{u}_t\|^2 \qquad \mathbf{g}_t = \nabla f|_t$$

- This turns out to be equivalent to the Poisson equation:

$$\begin{aligned} \nabla \cdot \nabla f &= \nabla \cdot u \quad \text{in } M \\ n \cdot \nabla f &= n \cdot u \quad \text{on } \partial M \end{aligned}$$

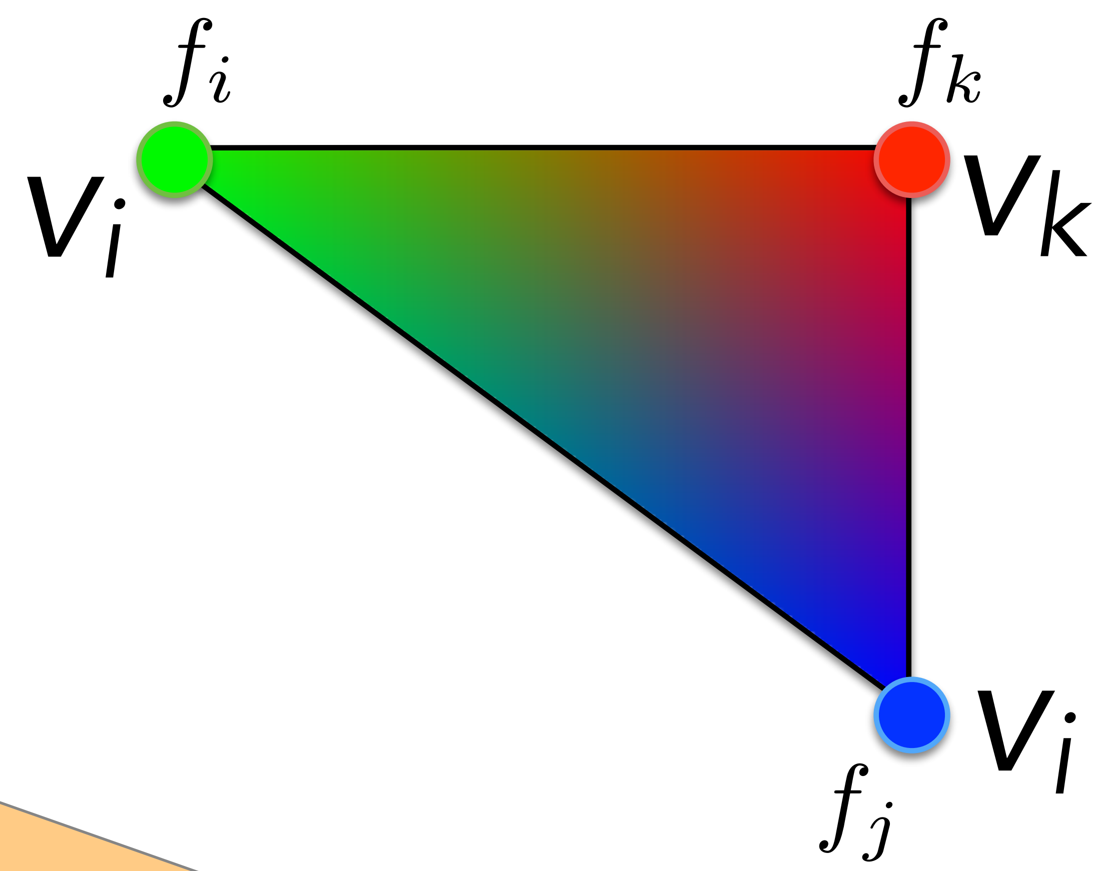
- Can also be written as a quadratic form:

$$\frac{1}{2}\mathbf{f}^T K \mathbf{f} + \mathbf{f}^T \mathbf{b} + c$$

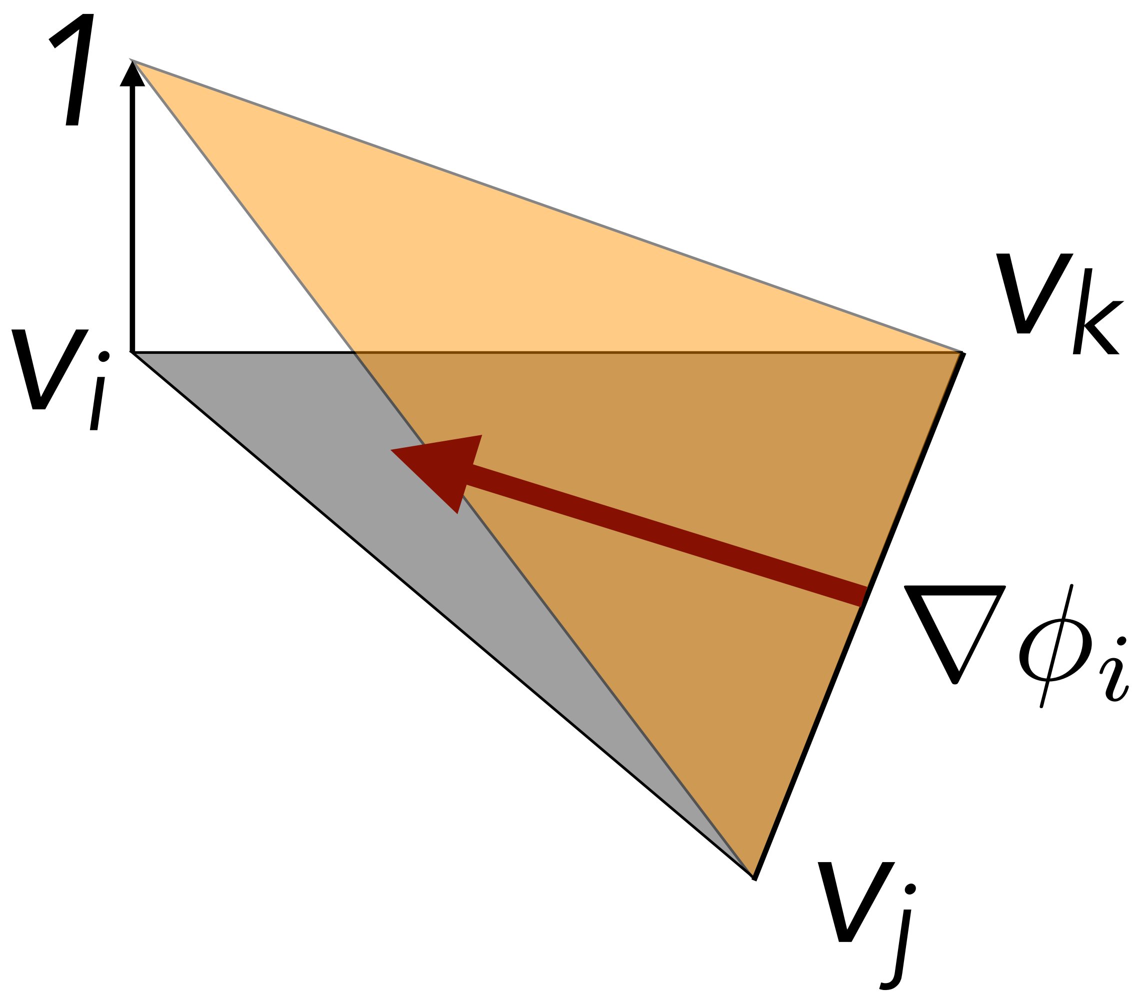
- Only determined up to constant: fix one vertex to zero!

# Expressing as Quadratic Form

$$\min \sum_{\text{face } t} w_t \|\mathbf{g}_t - \mathbf{u}_t\|^2 \qquad \mathbf{g}_t = \nabla f|_t$$




- Recall, piecewise linear f can be written as a sum of "hat" basis functions:




$$f(x) = f_i \phi_i(x) + f_j \phi_j(x) + f_k \phi_k(x)$$

$$\nabla f(x) = f_i \nabla \phi_i(x) + f_j \nabla \phi_j(x) + f_k \nabla \phi_k(x)$$

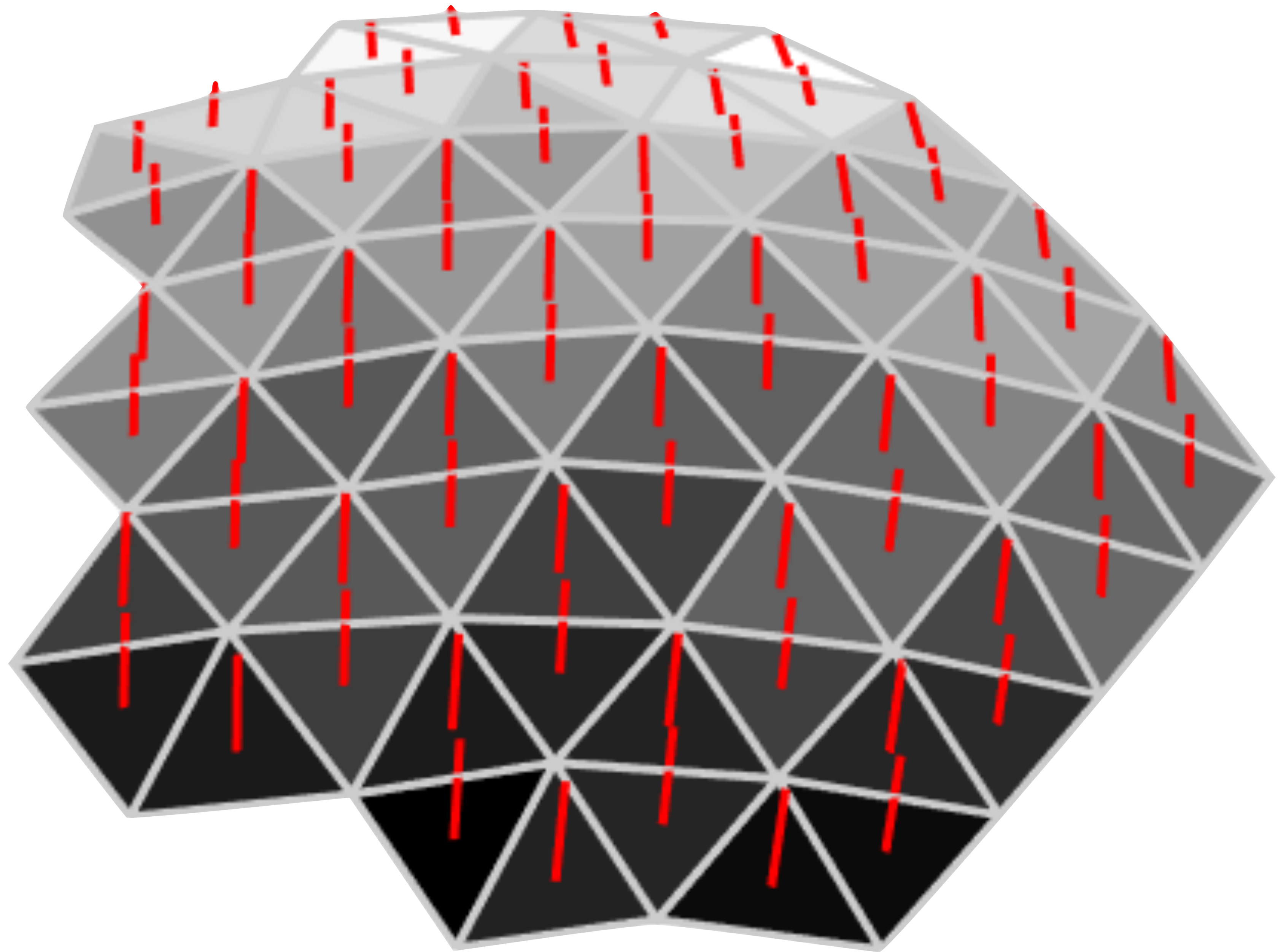
# The "Gradient Matrix"

- So the gradient is just a linear combination of vertex scalar field values

$$\underbrace{\begin{bmatrix} \left.\frac{\partial f}{\partial x}\right|_{t_1} \\ \vdots \\ \left.\frac{\partial f}{\partial x}\right|_{t_{\#F}} \\ \left.\frac{\partial f}{\partial y}\right|_{t_1} \\ \vdots \\ \left.\frac{\partial f}{\partial z}\right|_{t_{\#F}} \end{bmatrix}}_{3\#F \times 1} = G \underbrace{\begin{bmatrix} f|_{v_1} \\ \vdots \\ f|_{v_{\#V}} \end{bmatrix}}_{\#V \times 1}$$

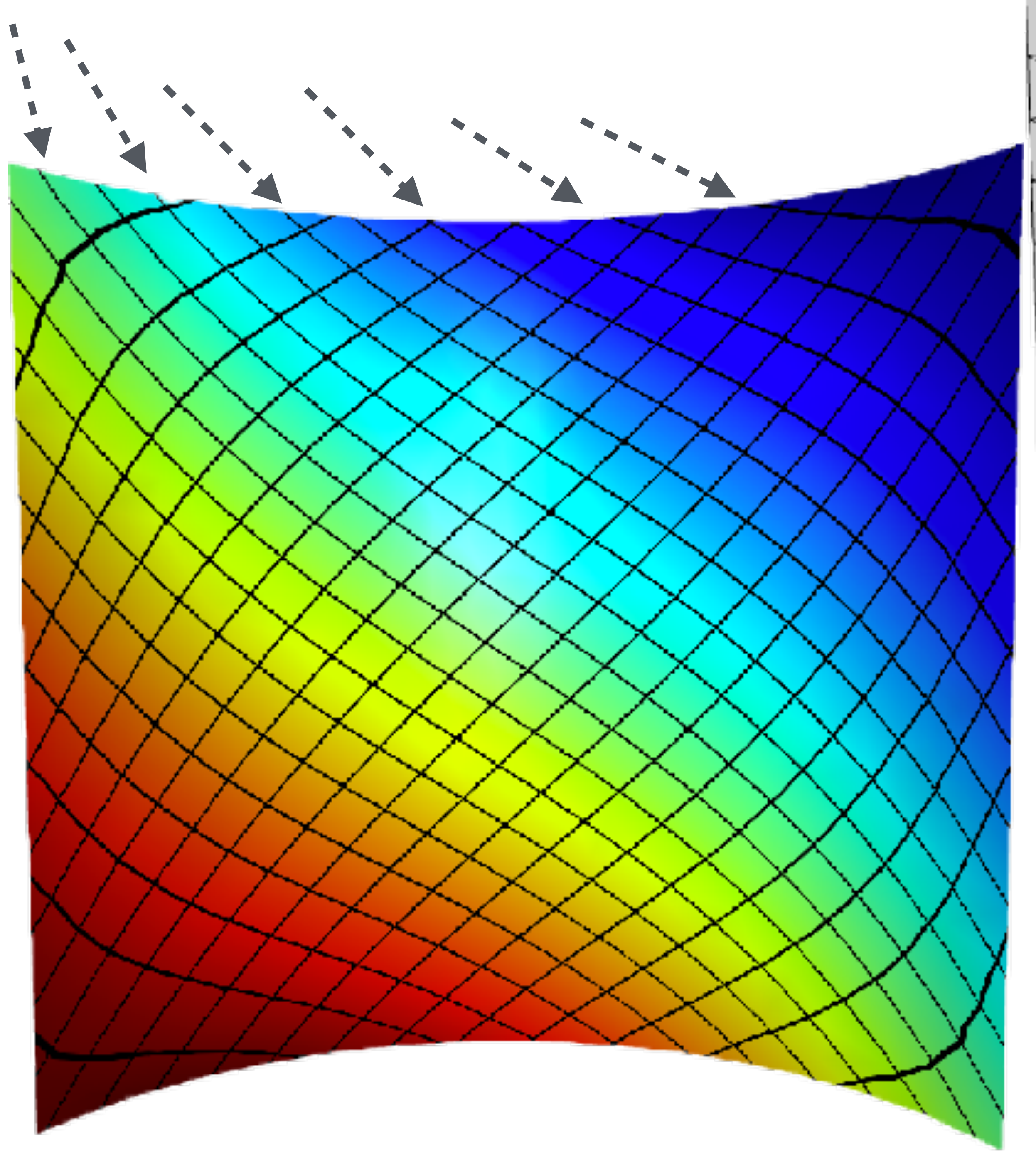
$$\underset{3\#F \times \#V}{G}$$

Maps per-vertex scalars to per-triangle gradient vectors

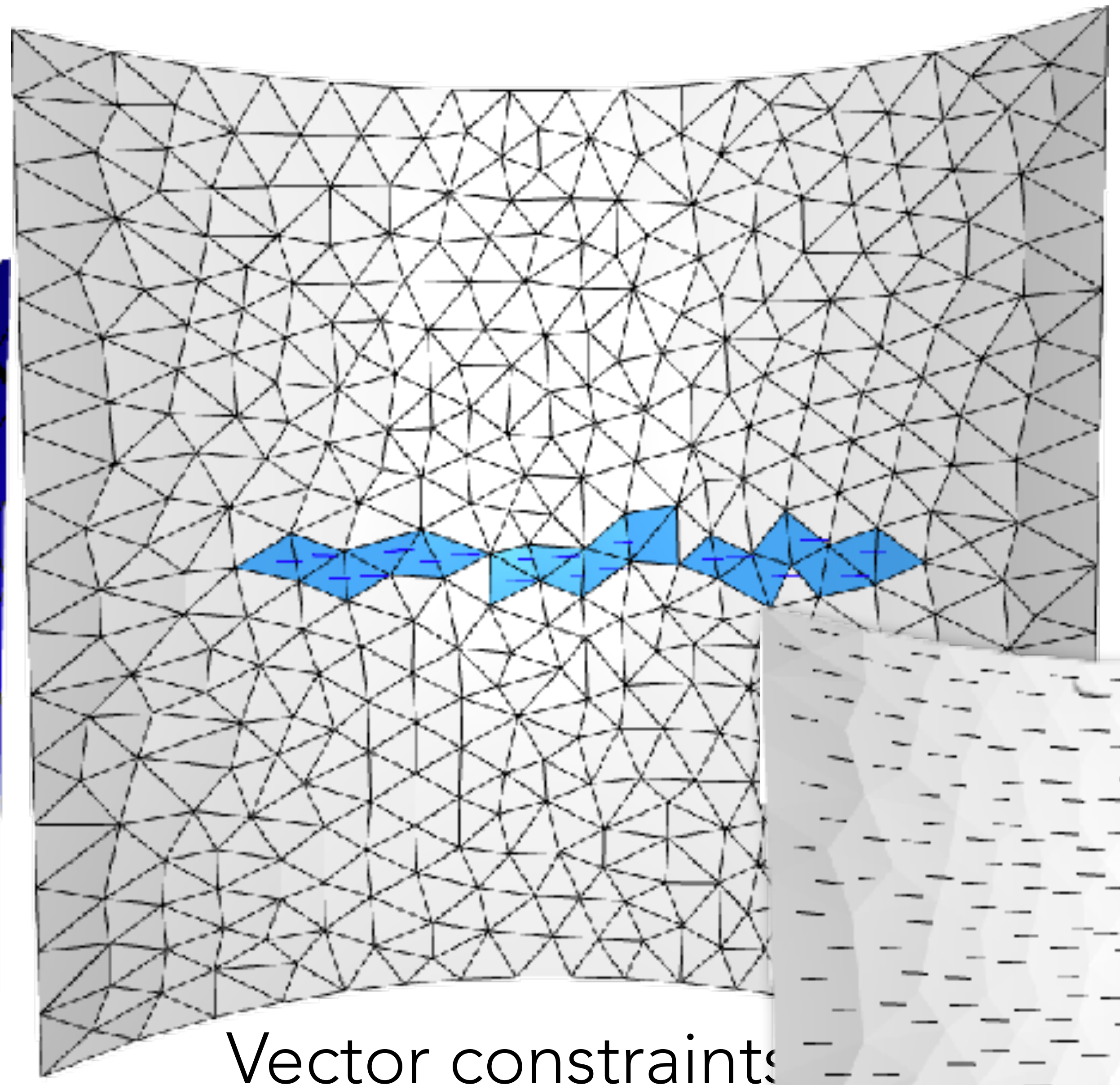


G = igl.grad(v, f)

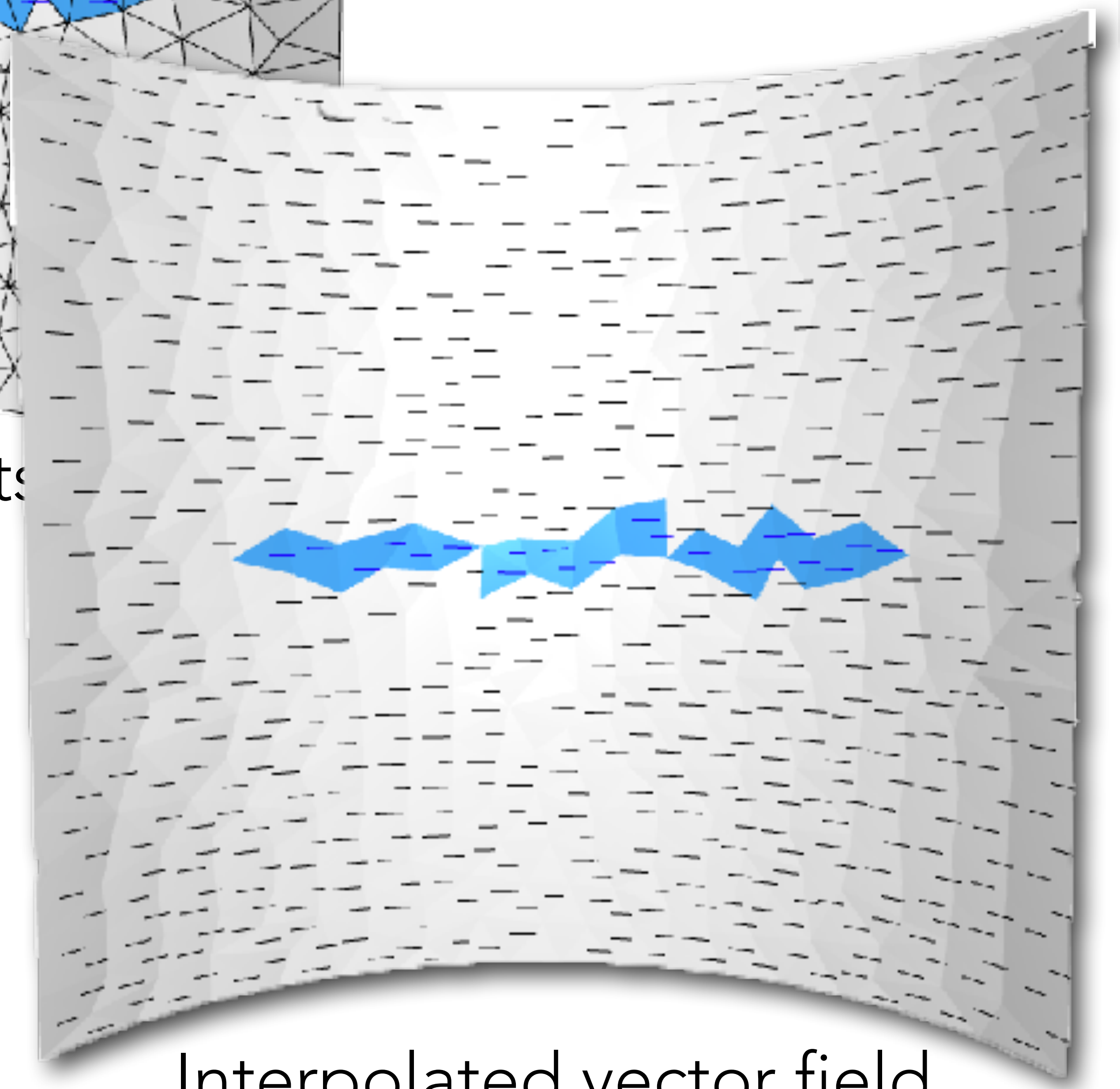
# Application: Edit Parameterizations



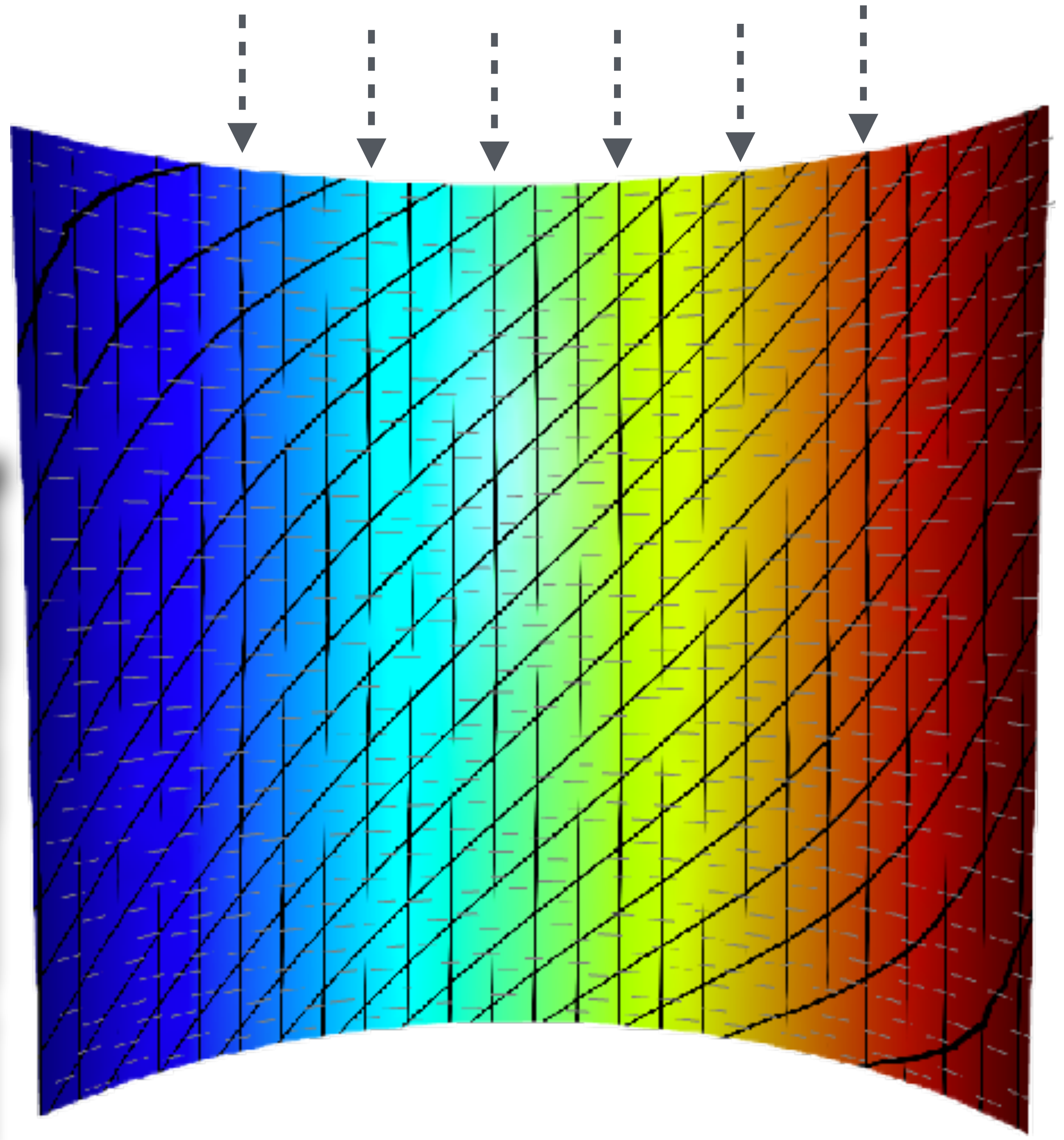
Harmonic parameterization



Vector constraints for u's gradient



Interpolated vector field (approximate u gradient)



Edited parameterization

# Equality Constraints

- Both steps involve equality constraints
  - Painted vectors on constrained faces
  - Scalar field value on single vertex fixed to zero
- Two approaches:
  - Lagrange multipliers (standard approach)
  - Variable elimination (much more efficient here)

# Variable Elimination via Row/Col Removal

- I'll derive this since it's less discussed than Lagrange multipliers

- Most efficient way to fix variables to constants:

$$\min_{\mathbf{x}} \frac{1}{2}\mathbf{x}^T A\mathbf{x} - \mathbf{b}^T\mathbf{x} + c \quad \text{s.t.} \quad \mathbf{x}_c = \mathbf{d}$$

$$\left(\mathbf{x} = \begin{bmatrix} \mathbf{x}_f \\ \mathbf{x}_c \end{bmatrix}\right)$$

Free Variables

Constrained Variables

- For derivation, assume vars ordered like this;
  Trick works for any ordering

Note that the matrix A is not the same A as in the code of the assignment.

# Row/Col Removal

$$
\begin{aligned}
&\frac{1}{2}\mathbf{x}^T A\mathbf{x} - \mathbf{b}^T\mathbf{x} + c \\
=&\frac{1}{2}\begin{bmatrix}\mathbf{x}_f^T & \mathbf{x}_c^T\end{bmatrix}\begin{bmatrix}A_{ff} & A_{fc} \\ A_{cf} & A_{cc}\end{bmatrix}\begin{bmatrix}\mathbf{x}_f \\ \mathbf{x}_c\end{bmatrix} - \begin{bmatrix}\mathbf{b}_f^T & \mathbf{b}_c^T\end{bmatrix}\begin{bmatrix}\mathbf{x}_f \\ \mathbf{x}_c\end{bmatrix} + c \\
=&\frac{1}{2}\left(\mathbf{x}_f^T A_{ff}\mathbf{x}_f + 2\mathbf{x}_f^T A_{fc}\mathbf{x}_c + \overset{\text{const}}{\cancel{\mathbf{x}_c^T A_{cc}\mathbf{x}_c}}\right) - \mathbf{b}_f^T\mathbf{x}_f - \overset{\text{const}}{\cancel{\mathbf{b}_c^T\mathbf{x}_c}} + c
\end{aligned}
$$

$$
\min_{\mathbf{x}_f} \quad \Longrightarrow \quad 0 = A_{ff}\mathbf{x}_f + A_{fc}\mathbf{x}_c - \mathbf{b}_f
$$

$$
\tilde{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}
$$

$$
\tilde{A} = A_{ff} \quad \tilde{\mathbf{x}} = \mathbf{x}_f \quad \tilde{\mathbf{b}} = \mathbf{b}_f - A_{fc}\mathbf{x}_c
$$

Rows/cols corresponding to fixed variables removed!

Columns corresponding to fixed vars contribute to RHS

# Questions?